



Technische
Universität
Dresden



Machine Learning I

Digitization and Data Analytics

Jana Klimpke

Structure of this Lecture

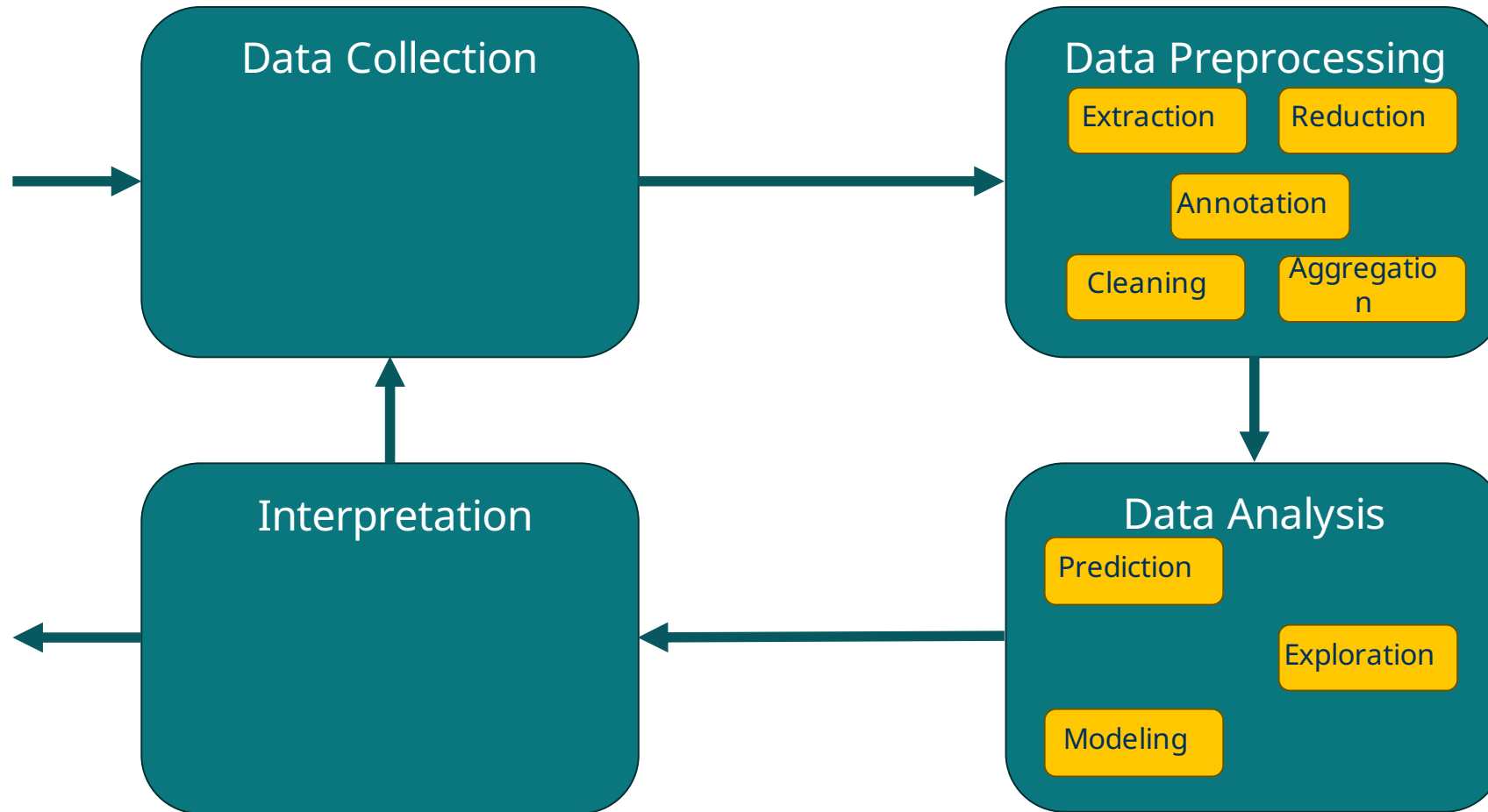
1. Introduction to Machine Learning

2. Similarity and Distances

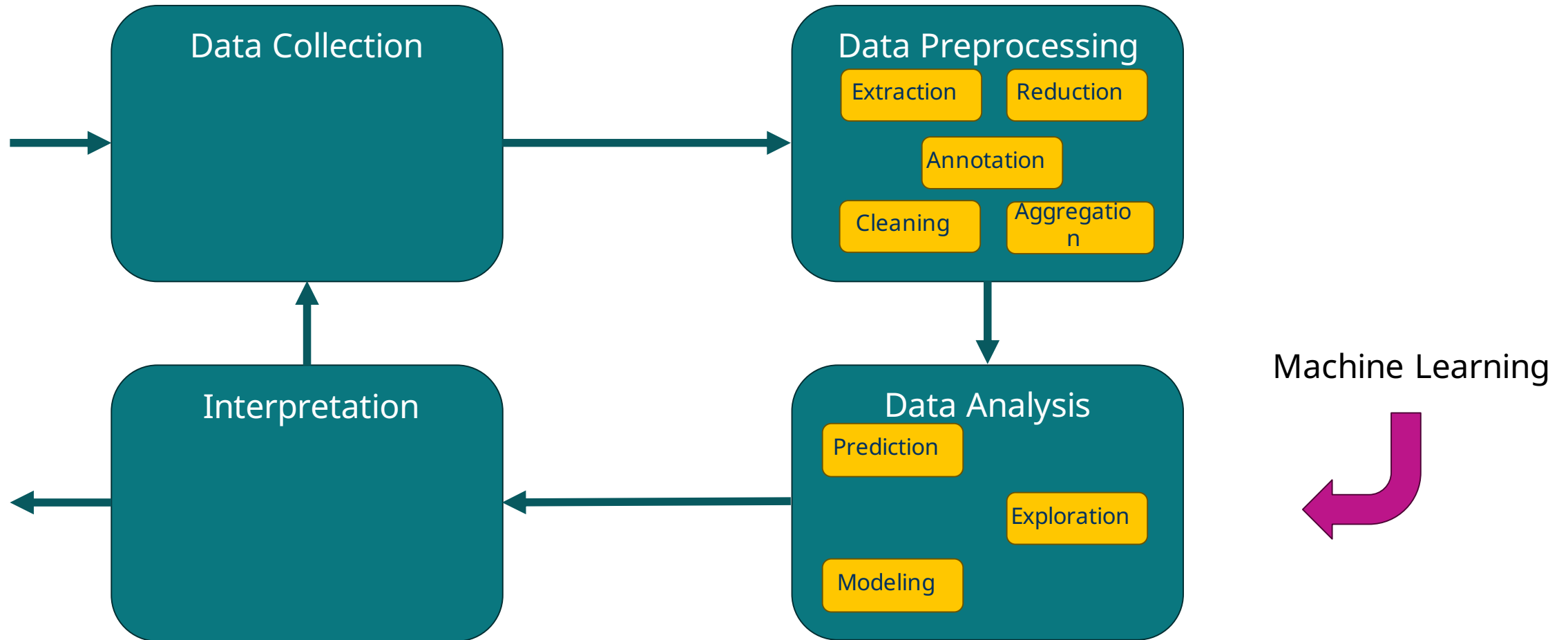
3. Clustering

- 3.1 Overview
- 3.2 Partitioning
- 3.3 Hierarchical clustering
- 3.4 Density-based
- 3.4 Cluster assessment
- 3.5 Aspects of big data clustering

Data Processing and Analytics



Data Processing and Analytics



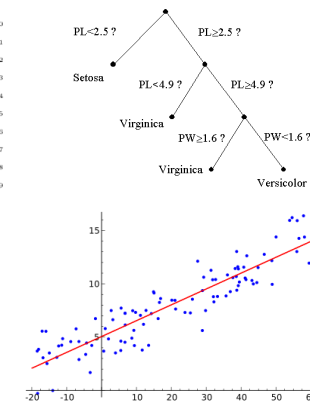
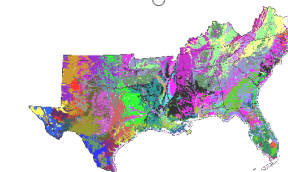
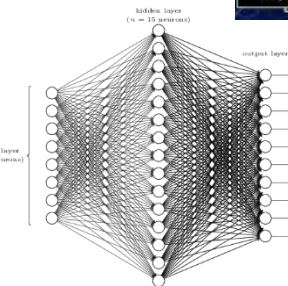
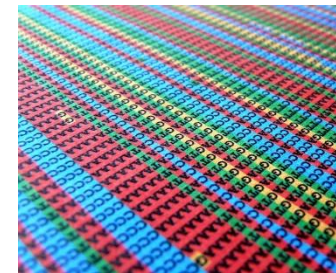
Data Analytics and Machine Learning

Goals of Data Analytics

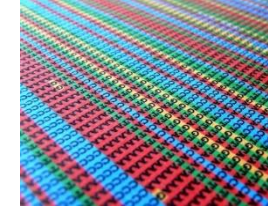
- Recognize complex potential patterns in existing data
- Support decision making

Based on

- Large amount of data
- Data analytics methods



Data Analytics and Machine Learning



- Big data analytics need automated methods
- Machine Learning provides automated methods of data analysis

„[...] we define machine learning as a set of methods that can automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other kinds of decision making under uncertainty [...]”

Murphy, K.P. Machine Learning A Probabilistic Perspective, MIT Press, 2012

Types of Machine Learning

Supervised Learning (Predictive Analysis)

Unsupervised Learning (Descriptive/Exploratory Analysis)

Semi-Supervised Learning

Reinforcement Learning

Supervised Learning (Predictive Analysis)

- Describes properties of future data
- Question „What could happen?“
- Goal: Predicting future
 - E.g. forecasting, scoring, classification of unknown data

Supervised Learning (Predictive Analysis)

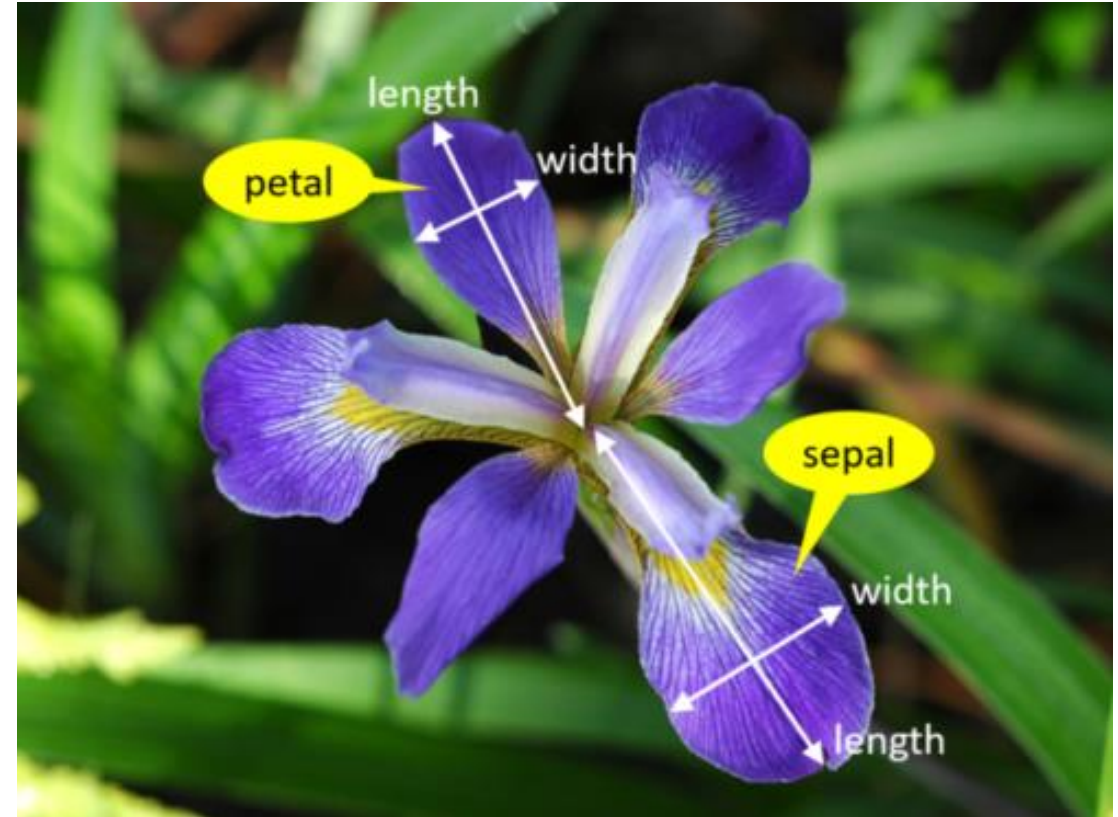
Labeled Data

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.9	3.1	1.5	0.1	setosa
5.1	3.8	1.5	0.3	setosa
4.7	3.2	1.6	0.2	setosa
5.1	3.4	1.5	0.2	setosa
5.0	3.3	1.4	0.2	setosa
5.2	2.7	3.9	1.4	versicolor
5.7	2.6	3.5	1.0	versicolor
5.7	2.8	4.1	1.3	versicolor
6.0	2.2	5.0	1.5	virginica

Courtesy of Iryna Okhrin

Iris flower data set, Ronald Fisher,

The use of multiple measurements in taxonomic problems, 1936



<https://www.integratedots.com/determine-number-of-iris-species-with-k-means/>

Supervised Learning (Predictive Analysis)

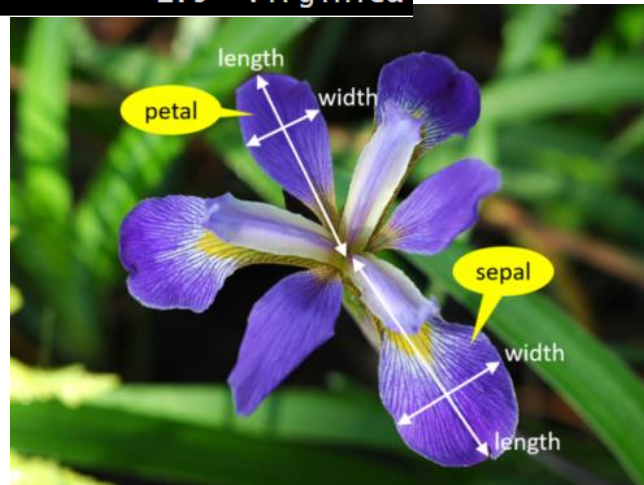
Labeled Data

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.9	3.1	1.5	0.1	setosa
5.1	3.8	1.5	0.3	setosa
4.7	3.2	1.6	0.2	setosa
5.1	3.4	1.5	0.2	setosa
5.0	3.3	1.4	0.2	setosa
5.2	2.7	3.9	1.4	versicolor
5.7	2.6	3.5	1.0	versicolor
5.7	2.8	4.1	1.3	versicolor
6.0	2.2	5.0	1.5	virginica

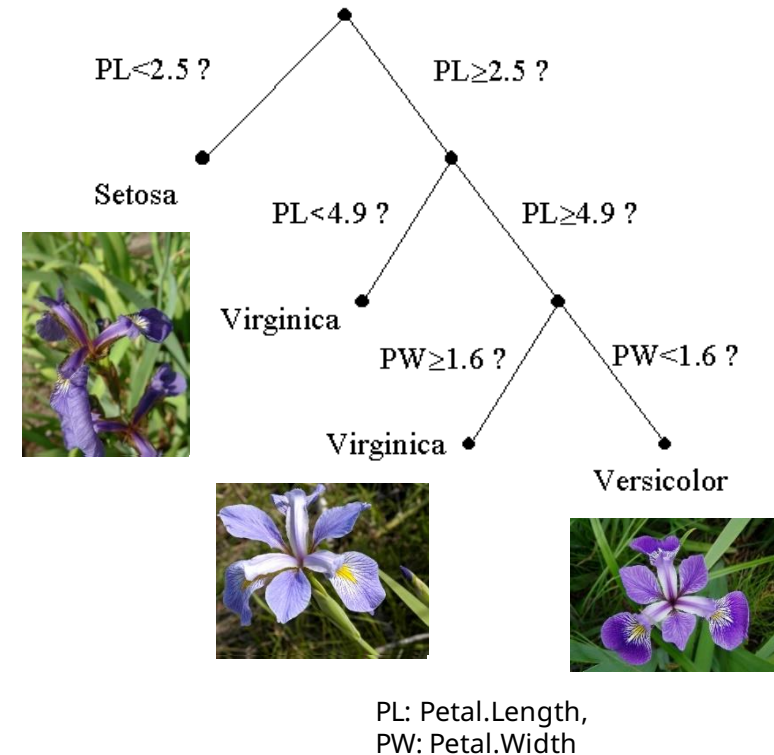
Courtesy of Iryna Okhrin

Iris flower data set, Ronald Fisher,

The use of multiple measurements in taxonomic problems, 1936



Goal: Classify unknown data



Supervised Learning - Classification

Given

- $D = \{(\vec{x}_j, \vec{y}_j)\}_{j=1}^N$ training set of labeled input-output pairs
- N training examples
 - $(\vec{x}, \vec{y})$ input-output pairs
 - x_i : feature (attribute, covariate, predictors)
 - y_i : label (response variable, target)

$\vec{x}$: feature vector

$\vec{y}$: output vector

supervised

Goal

- Learn mapping $f: \vec{x} \rightarrow \vec{y}$

Supervised Learning - Classification

- Informally:
 - Describes properties of future data
 - Question „What could happen?“
 - Goal: Predicting future
 - E.g. forecasting, scoring, classification of unknown data
- Based on formal description:
 - learn (fit) a model that relates the label (response) to the features (predictors)
 - accurately predict the label for future observations
 - better understanding the relationship between the labels and the features
- Probabilistic view:
 - fitting a conditional model, $p(y|x)$, which specifies (a distribution over) outputs given inputs

Unsupervised Learning (Descriptive/Exploratory Analysis)

- Describes existing data
- Question „What has happened?“
- Goal: Extract potential patterns

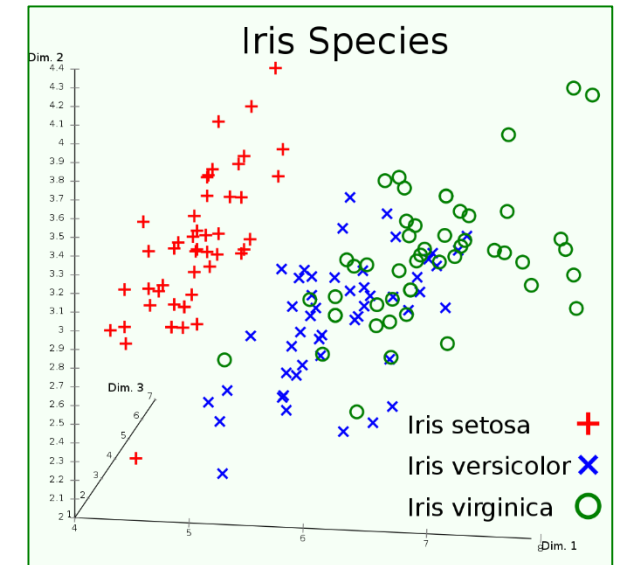
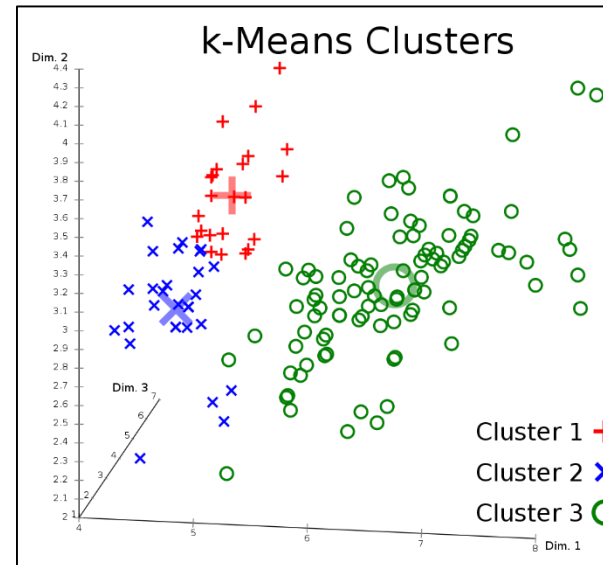
Unsupervised Learning (Descriptive/Exploratory Analysis)

Goal: **Discover groups** within data

Unlabeled data

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.9	3.1	1.5	0.1	setosa
5.1	3.8	1.5	0.3	setosa
4.7	3.2	1.6	0.2	setosa
5.1	3.4	1.5	0.2	setosa
5.0	3.3	1.4	0.2	setosa
5.2	2.7	3.9	1.4	versicolor
5.7	2.6	3.5	1.0	versicolor
5.7	2.8	4.1	1.3	versicolor
6.0	2.2	5.0	1.5	virginica

Courtesy of Iryna Okhrin



Learned model does **not** include **semantic meaning** of clusters

Iris flower data set, Ronald Fisher, *The use of multiple measurements in taxonomic problems*, 1936

Unsupervised Learning (Descriptive/Exploratory Analysis)

- Given

- $D = \{(\vec{x}_j)\}_{j=1}^N$ training set,
- N training examples
- $(\vec{x}_j)$ inputs

- x_i : feature (attribute, covariate, predictors) $\vec{x}$: feature vector

- Goal

- Find interesting patterns

Unsupervised Learning (Descriptive/Exploratory Analysis)

Informally

- Describes existing data
- Question „What has happened?“
- Goal: Extract potential patterns

Based on formal description

- Group data points $\vec{x}_j$, based on an application dependent distance measure

Probabilistic view

- fitting an unconditional model of the form $p(x)$, which can generate new data x

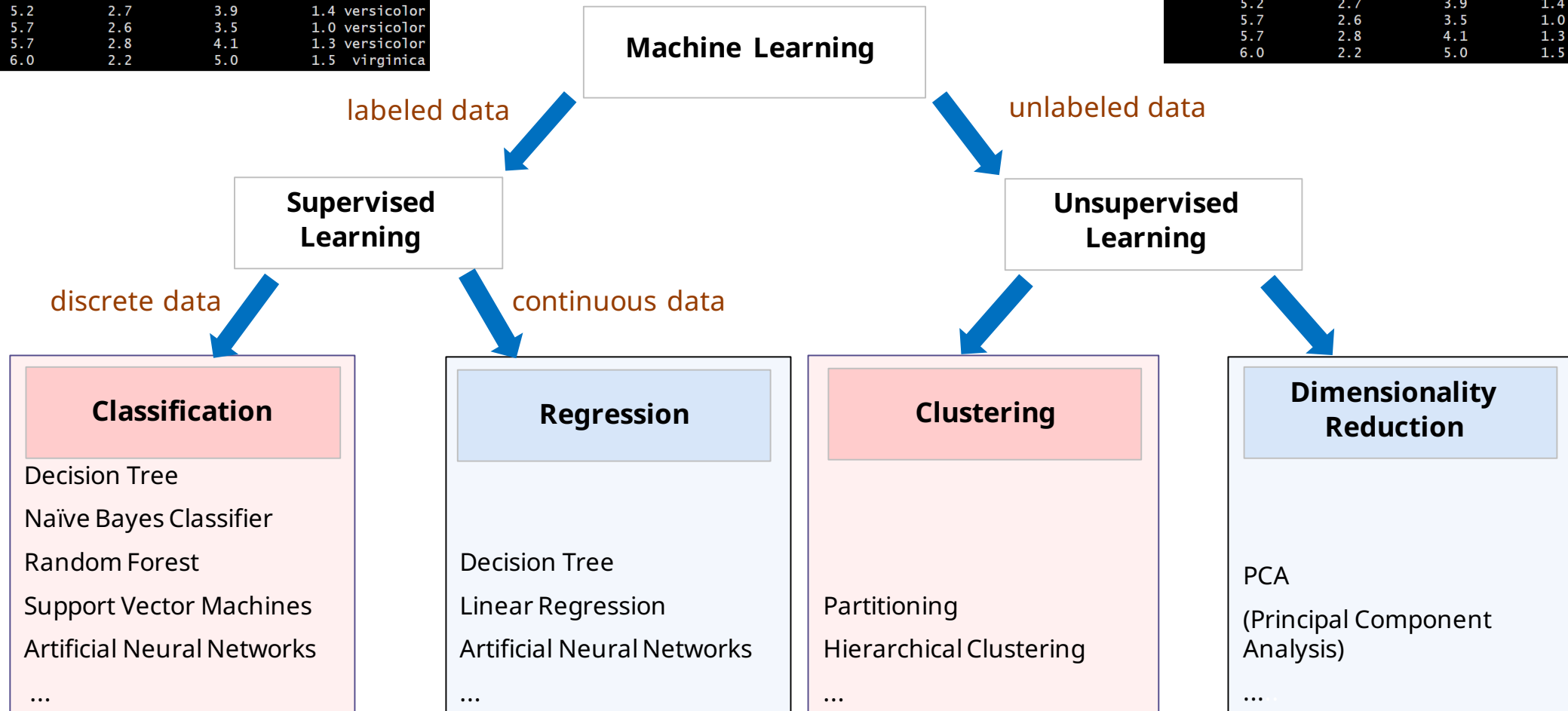
Challenge

- What kind of pattern is interesting and should be searched for?
- How to assess the extracted pattern?

Machine Learning – Takeaway

Sepal.Length	Sepal.width	Petal.Length	Petal.Width	Species
4.9	3.1	1.5	0.1	setosa
5.1	3.8	1.5	0.3	setosa
4.7	3.2	1.6	0.2	setosa
5.1	3.4	1.5	0.2	setosa
5.0	3.3	1.4	0.2	setosa
5.2	2.7	3.9	1.4	versicolor
5.7	2.6	3.5	1.0	versicolor
5.7	2.8	4.1	1.3	versicolor
6.0	2.2	5.0	1.5	virginica

Sepal.Length	Sepal.width	Petal.Length	Petal.Width
4.9	3.1	1.5	0.1
5.1	3.8	1.5	0.3
4.7	3.2	1.6	0.2
5.1	3.4	1.5	0.2
5.0	3.3	1.4	0.2
5.2	2.7	3.9	1.4
5.7	2.6	3.5	1.0
5.7	2.8	4.1	1.3
6.0	2.2	5.0	1.5



Structure of this Lecture

1. Introduction to Machine Learning

2. Similarity and Distances

3. Clustering

- 3.1 Overview
- 3.2 Partitioning
- 3.3 Hierarchical clustering
- 3.4 Density-based
- 3.4 Cluster assessment
- 3.5 Aspects of big data clustering

Similarity and Distances

Requirements for many Machine Learning applications

- determination of similar or dissimilar objects, patterns, attributes, and events
- computation of similarity



Formal statement

- Given two objects ***O1*** and ***O2***, determine a value of the similarity ***Sim(O1 , O2)*** (or distance ***Dist(O1 , O2)***) between the two objects.

Possible Objects

- Images
- Text Documents
- Time Series
- Mixed data
- ...

Similarity and Distances

Question:

What's **more similar** - **green** and **blue** or **green** and **red**?

Answer:

green and **blue**

Why?

abcdefghijklmnopqrstuvwxyz

a**b**cdef**g**hijklmnopq**r**stuvwxyz

*„Poor choices of the **distance function** can sometimes be disastrously misleading depending on the **application domain**.“*

[Charu C. Aggarwal, *“Data Mining The Textbook”*]

Similarity and Distances

- The terms „*distance function*“, „*similarity function*“, „*dissimilarity function*“ are used depending on the application domain
- Similarity
 - Numerical measure of **how alike** two data objects are
 - Value is higher when objects are more alike
 - **Often** falls in the **range [0,1]**
- Dissimilarity (e.g., distance)
 - Numerical measure of **how different** two data objects are
 - Lower when objects are more alike
 - **Minimum** dissimilarity is **often 0**
 - **Upper limit varies**

[Similarity and Distances see Aggarwal, C., Data Mining Textbook, chapter 3.2]

Similarity and Distances – Multidimensional Data

Quantitative Data

$$\bar{x} = (x_1, \dots, x_d)$$

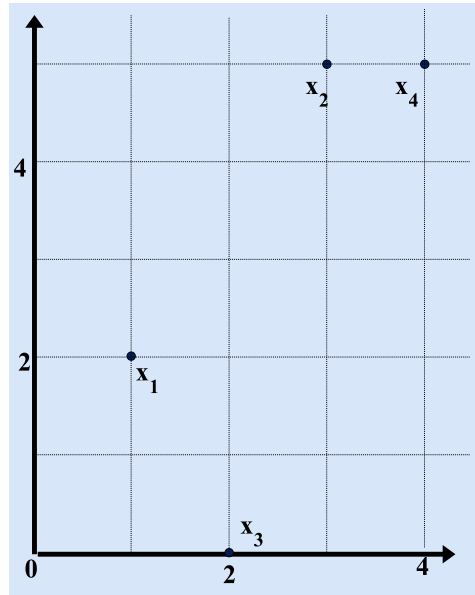
- Distance: L_p -Norm

$$Dist(\bar{x}, \bar{y}) = \left(\sum_{i=1}^d |x_i - y_i|^p \right)^{\frac{1}{p}}$$

- L_1 -Norm: Manhattan
- L_2 -Norm: Euclidean
- L_∞ -Norm: Maximum Norm

Example: L_p -Norm (Minkowski Distance)

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Euclidean (L_2)

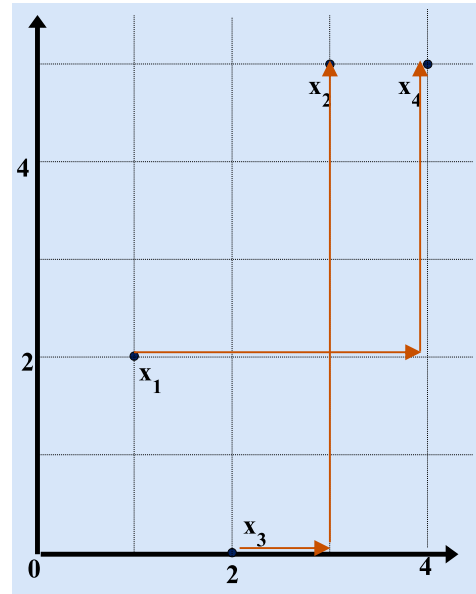
L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

Maximum (L_∞)

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

Example: L_p -Norm (Minkowski Distance)

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

$$Dist(\vec{x}, \vec{y}) = \left(\sum_{i=1}^d |x_i - y_i| \right)$$

Euclidean (L_2)

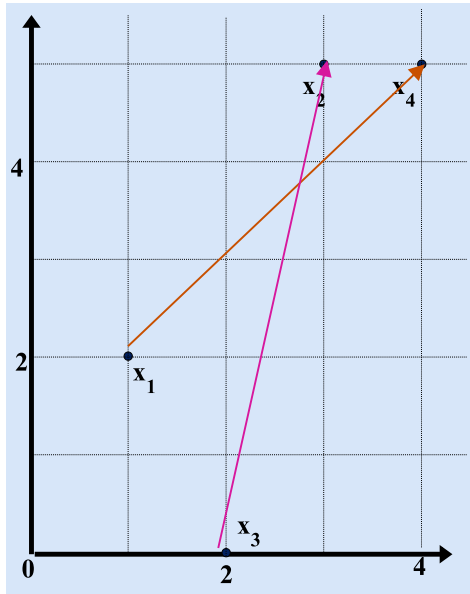
L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

Maximum (L_∞)

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

Example: L_p -Norm (Minkowski Distance)

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Euclidean (L_2)

L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

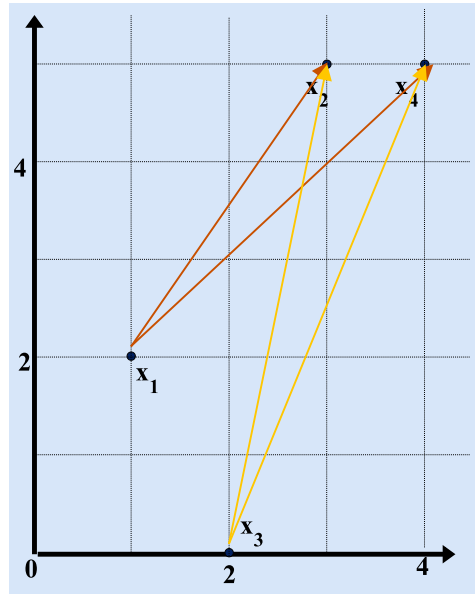
$$Dist(\vec{x}, \vec{y}) = \left(\sum_{i=1}^d |x_i - y_i|^2 \right)^{\frac{1}{2}}$$

Maximum (L_∞)

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

Example: L_p -Norm (Minkowski Distance)

point	attribute 1	attribute 2
x1	1	2
x2	3	5
x3	2	0
x4	4	5



Manhattan (L_1)

L	x1	x2	x3	x4
x1	0			
x2	5	0		
x3	3	6	0	
x4	6	1	7	0

Euclidean (L_2)

L2	x1	x2	x3	x4
x1	0			
x2	3.61	0		
x3	2.24	5.1	0	
x4	4.24	1	5.39	0

Maximum (L_∞)

L_∞	x1	x2	x3	x4
x1	0			
x2	3	0		
x3	2	5	0	
x4	3	1	5	0

$$Dist(\vec{x}, \vec{y}) = \max_{i=1, \dots, d} |x_i - y_i|$$

Similarity and Distances – Multidimensional Data

Categorical Data

no ordering among discrete values of categorical data

base **similarity** on similarities of **individual attribute values** $S(x_i, y_i)$

$$\bar{x} = (x_1, \dots, x_d)$$
$$\bar{y} = (y_1, \dots, y_d)$$

- $S(\bar{x}, \bar{y}) = \sum_{i=1}^d S(x_i, y_i)$

Simplest $S(x_i, y_i)$:

- $S(x_i, y_i) = \begin{cases} 1, & x_i = y_i \\ 0, & x_i \neq y_i \end{cases}$

ID	Last Name	First Name	Adult	...
1	Meyer	Paul	1	
2	Smith	Jane	1	
3	Miller	Mike	1	
4	Barber	Susan	0	
5	Green	Alice	1	
6	Farmer	Paul	0	

Similarity and Distances – Multidimensional Data

Categorical Data

no ordering among discrete values of categorical data

base **similarity** on similarities of **individual attribute values** $S(x_i, y_i)$

$$\begin{aligned}\bar{x} &= (x_1, \dots, x_d) \\ \bar{y} &= (y_1, \dots, y_d)\end{aligned}$$

$$S(\bar{x}, \bar{y}) = \sum_{i=1}^d S(x_i, y_i)$$

Inverse Occurrence Frequency

$$p_k(\mathbf{x}) = \frac{\# \text{ datapoints } \bar{x} | k\text{th attribute} = x_k}{\# \text{ datapoints } \bar{x}}$$

$$S(x_i, y_i) = \begin{cases} 1/p_k(x_i)^2, & x_i = y_i \\ 0 & x_i \neq y_i \end{cases}$$

ID	Last Name	First Name	Adult	...
1	Meyer	Paul	1	
2	Smith	Jane	1	
3	Miller	Mike	1	
4	Barber	Susan	0	
5	Green	Alice	1	
6	Farmer	Paul	0	

$$p_{Adult}(0) = \frac{1}{3}, \quad p_{Adult}(1) = \frac{2}{3}$$

Similarity and Distances – Text Data

Cosine Similarity

- Computes angle between two document vectors
- Insensitive to absolute length of document
- Simply uses raw frequencies

C	doc1	doc2	doc3	doc4	doc5	doc6
ship	1	0	1	0	0	0
boat	0	1	0	0	0	0
ocean	1	1	0	0	0	0
voyage	1	0	0	1	1	0
trip	0	0	0	1	0	1

$$\cos(\bar{x}, \bar{y}) = \frac{\sum_{i=1}^d x_i y_i}{\sqrt{\sum_{i=1}^d x_i^2} \sqrt{\sum_{i=1}^d y_i^2}}$$

$$\begin{aligned}\bar{x} &= (x_1, \dots, x_d) \\ \bar{y} &= (y_1, \dots, y_d)\end{aligned}$$

Applications: information retrieval, biologic taxonomy, gene feature mapping, ...

Similarity and Distances – Discrete Sequences

Data

- Sequences of equal length
- categorical values

Edit Distance (Levenshtein distance)

- Least amount of cost to transform sequence into another
- Use insertions, deletions, replacements with specific costs, distance depends on costs

ababababab

$a \mapsto b, b \mapsto a$

bababababa Costs: 10 replacements

ababababab

delete a left,
insert a right

bababababa Costs: 1 insertions, 1 deletion

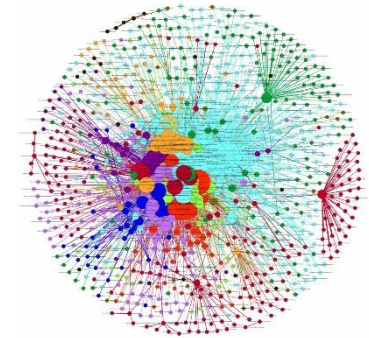
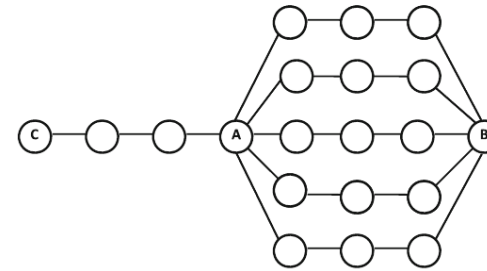
Longest Common Subsequence (LCSS)

- Sequences: xmjyauz mzjawxu
- Subsequences: xmjyauz mzjawxu (LCSS: mjau), LCSS can be computed in PTime

Similarity – Graph Similarity Measures

Similarity **between nodes**, based on connectivity

- Shortest paths
 - Dijkstra algorithm
- Many paths
 - PageRank
 - SimRank



Similarity **between two graphs**

- very challenging
- graph-isomorphism problem is NP-hard (identical graphs)
- Similarities and distances:
 - maximum common subgraph distance
 - substructure-based similarity
 - graph-edit distance

Similarity – Time Series

Equal length: L_p -Norm

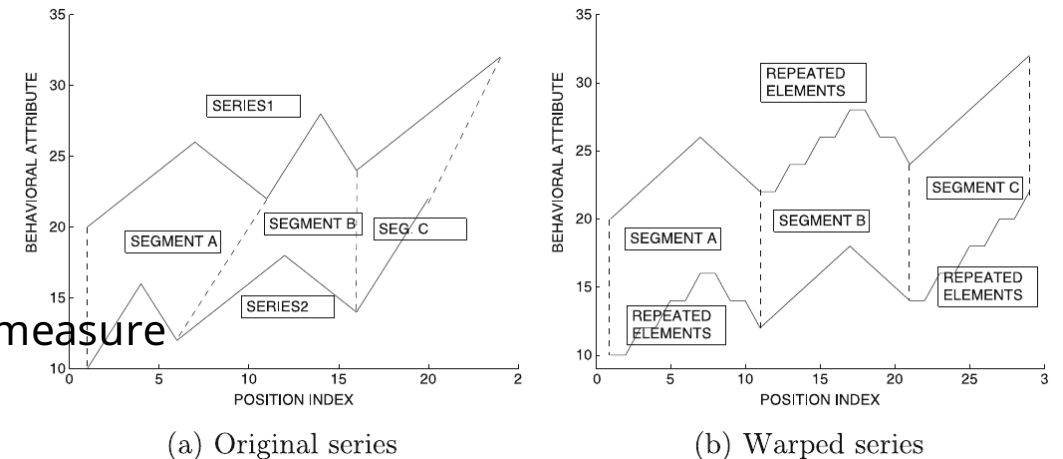
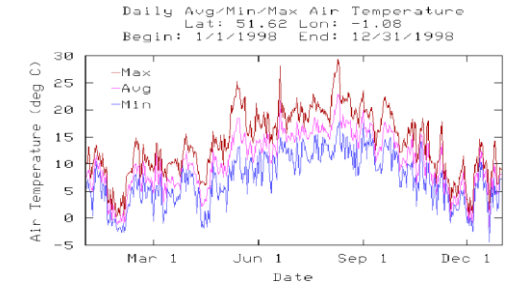
Problems:

- time series are **not** drawn on **same scale** (behavioral attribute **scaling** and **translation**)
- **absolute** time stamp is **not important**, series need to be **shifted** (temporal attribute translation)
- time series have **noisy** segments
- scale of temporal axis differs (temporal attribute scaling)

Dynamic Time Warping Distance (DTW)

- DTW **stretches** series along the **time axis**
- repeats some of the elements of either of the two series
- generated series of same length allow **apply any distance measure**

$$\bar{x} = (x_1, \dots, x_d)$$
$$\bar{y} = (y_1, \dots, y_d)$$



[Aggarwal, C., Data Mining Textbook]

Similarity and Distances – High Dimensional Data

High dimensional data

- $n \ll d$ (e.g. $n=100$ observations in $d=10^6$ dimensions)
- occurs in many applications as e.g. genome analysis, spatial statistics

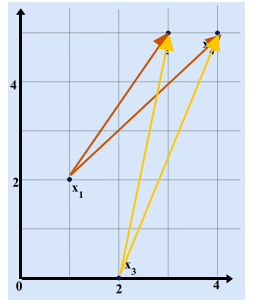
Curse of Dimensionality

- center of high dimensional spaces is almost empty
- points get scattered to the corners and boundary regions
- majority of vectors are orthogonal
- projection on lower space may cause severe loss of information

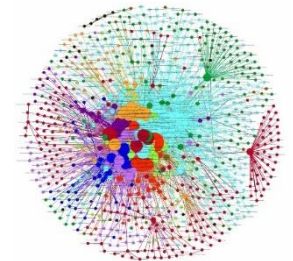
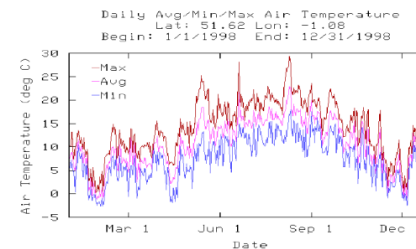
For more details see Zaki, M., Wagner M. Data Mining and Machine Learning: Fundamental Concepts and Algorithms, Cambridge University Press, 2020 <https://dataminingbook.info/videos/> „Sep 20, High Dimensional Data“.

Similarity and Distances – Takeaway

- Choice of similarity and distance measure essential for successful ML applications
- Can lead to very different results
- Similarity and distance measure data dependent
- Similarity and distance measure application dependent
- High dimensional data are different



C	doc1	doc2	doc3	doc4	doc5	doc6
ship	1	0	1	0	0	0
boat	0	1	0	0	0	0
ocean	1	1	0	0	0	0
voyage	1	0	0	1	1	0
trip	0	0	0	1	0	1



Structure of this Lecture

1. Introduction to Machine Learning

2. Similarity and Distances

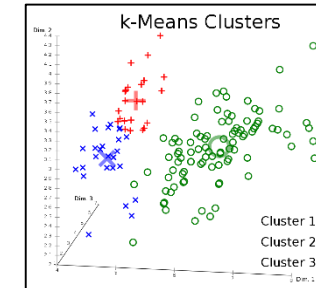
3. **Clustering**

- 3.1 Overview
- 3.2 Partitioning
- 3.3 Hierarchical clustering
- 3.4 Density-based
- 3.4 Cluster assessment
- 3.5 Aspects of big data clustering

Clustering

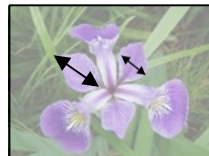
Cluster: A collection of data objects

- similar (or related) to one another **within** the **same** group
- dissimilar (or unrelated) to the **objects** in **other** groups



Clustering (cluster analysis, data segmentation)

- Finding **similarities** between data according to the characteristics found in the data
- **Grouping similar** data objects into clusters



Sepal.Length	Sepal.width	Petal.Length	Petal.Width
4.9	3.1	1.5	0.1
5.1	3.8	1.5	0.3
4.7	3.2	1.6	0.2
5.1	3.4	1.5	0.2
5.0	3.3	1.4	0.2
5.2	2.7	3.9	1.4
5.7	2.6	3.5	1.0
5.7	2.8	4.1	1.3
6.0	2.2	5.0	1.5

Clustering - Applications

- As a stand-alone tool to get insight into data distribution
- As a preprocessing step for other algorithms (e.g. outlier detection)

Clustering - Application Domains

Biology

- Taxonomy of living things: kingdom, phylum, class, order, family, genus and species

Information retrieval

- Document clustering

Land use

- Identification of areas of similar land use in an earth observation database

Marketing

- Help marketers discover distinct groups in their customer bases
- Use this knowledge to develop targeted marketing programs

City-planning

- Identifying groups of houses according to their house type, value, and geographical location

Climate

- Understanding earth climate, find patterns of atmospheric and ocean

Economic Science

- Market research

Considerations for Clustering

Partitioning criteria

- **Single** level vs. **hierarchical partitioning** (often, multi-level hierarchical partitioning is desirable)

Separation of clusters

- **Exclusive** (e.g., one customer belongs to only one region) vs. **non-exclusive** (e.g., one document may belong to more than one class)

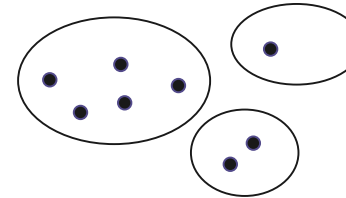
Similarity measure

- **Distance-based** (e.g., Euclidian, road network, vector) vs. **connectivity-based** (e.g., density or contiguity)

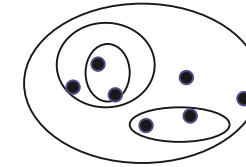
Clustering space

- **Full space** (often if low dimensional) vs. **subspaces** (often in high-dimensional clustering)

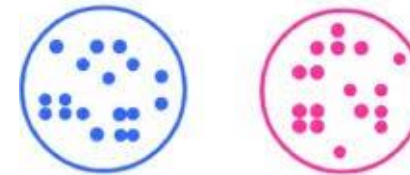
Single level partitioning



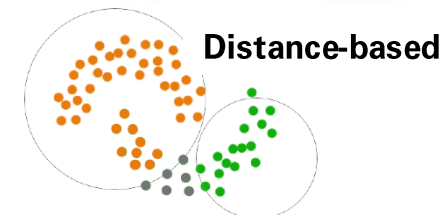
Hierarchical partitioning



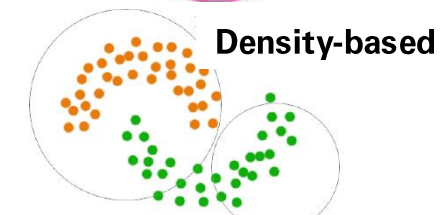
Exclusive Clustering



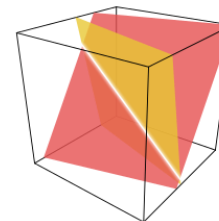
Overlapping Clustering



Distance-based



Density-based



Quality: What is a good Clustering?

Precondition

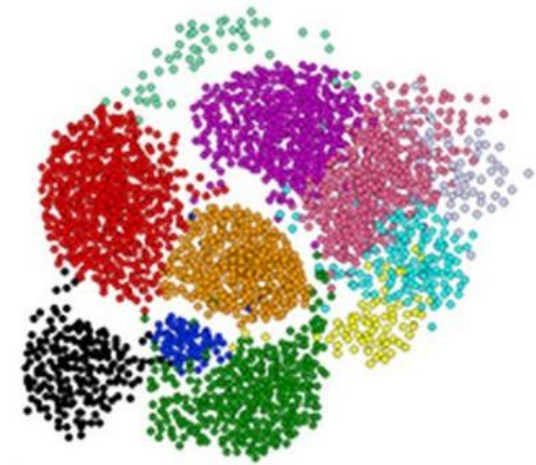
- **Existing patterns** within the data
- Does the application gives hints for such a **hypothesis**?

A **good clustering** method will produce high quality clusters

- **high intra-class similarity**: **cohesive** within clusters
- **low inter-class similarity**: **distinctive** between clusters

The **quality** of a clustering method **depends on**

- the **similarity measure** used by the method
- its **implementation**
- its ability to **discover** some or all of the **hidden patterns**



Clustering - A Hard Problem - Why?

- Clustering in two dimensions looks easy
- Clustering small amounts of data looks easy
- And in most cases, looks are not deceiving
- But: Many applications involve not 2, but 10 or 10,000 dimensions
- High-dimensional spaces look different:
- Almost all pairs of points are at about the same distance
- Almost any two vectors are almost orthogonal
- “Curse of Dimensionality”

J. Leskovec, A. Rajaraman, J. Ullman: Mining of Massive Datasets,
<http://www.mmids.org>

Major Clustering Approaches

- Partitioning approach
 - **Construct** various partitions and then **evaluate** them by some criterion, e.g., **minimizing** the sum of square errors
 - Typical methods: k-means, k-medoids, CLARANS
- **Hierarchical** approach
 - Create a hierarchical decomposition of the set of data (or objects) using some criterion
 - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- Density-based approach
 - Based on **connectivity** and **density** functions
 - Typical methods: DBSCAN, OPTICS, DenClue
- Grid-based approach
 - Based on a multiple-level granularity structure
 - Typical methods: STING, WaveCluster, CLIQUE

Structure of this Lecture

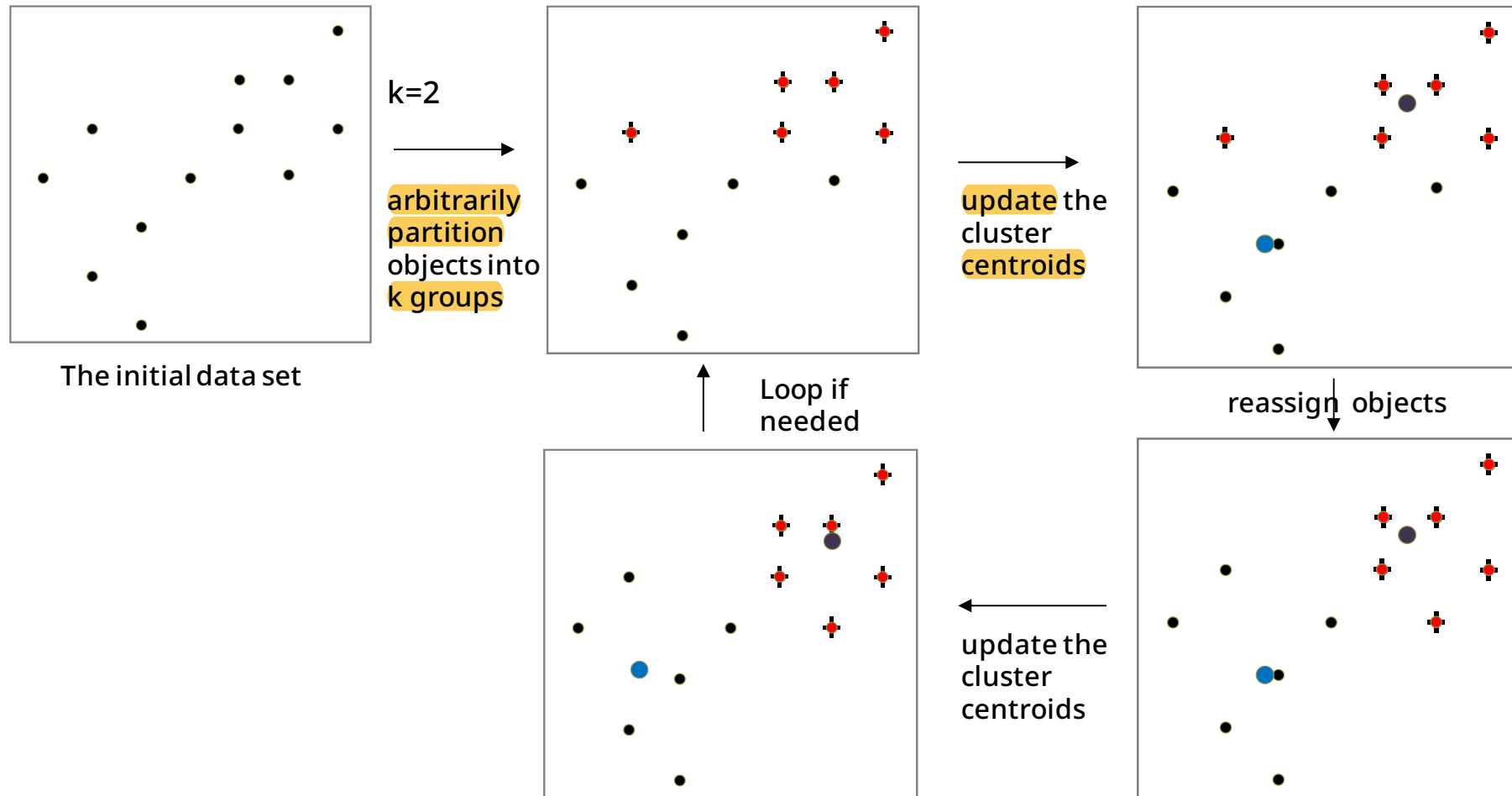
1. Introduction to Machine Learning

2. Similarity and Distances

3. Clustering

- 3.1 Overview
- 3.2 **Partitioning**
- 3.3 Hierarchical clustering
- 3.4 Density-based
- 3.4 Cluster assessment
- 3.5 Aspects of big data clustering

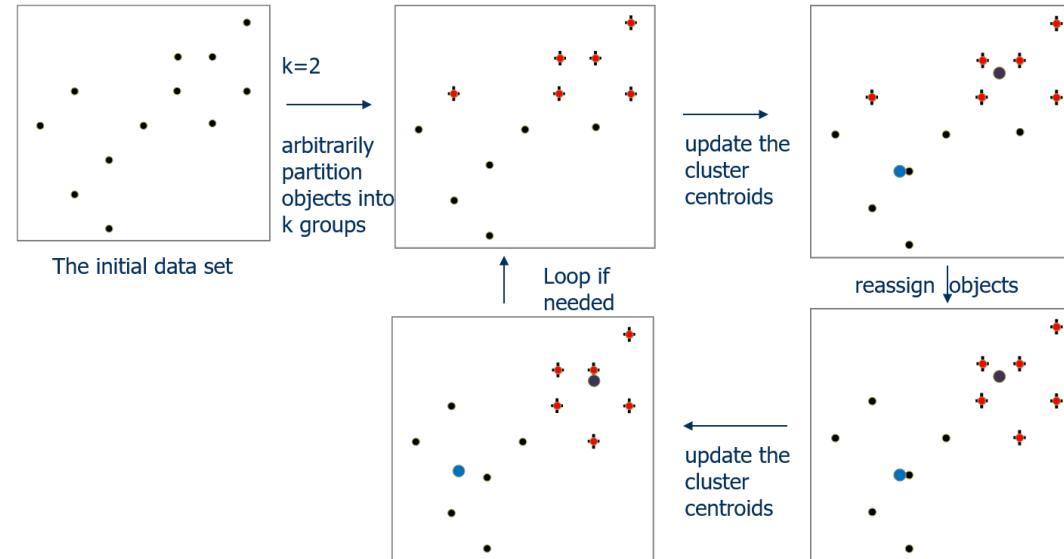
Partitioning – The k-Means Algorithm



Partitioning – The k-Means Algorithm

Algorithm:

- Partition objects into k nonempty subsets
- Repeat
 - Compute centroid (i.e., mean point) for each partition
 - Assign each object to the cluster of its nearest centroid
- Until no change



Lloyd, Stuart P. "Least squares quantization in PCM." Information Theory, IEEE Transactions on 28.2 (1982): 129-137.

Partitioning Methods

Partitioning

- A set $X = \{x_1, \dots, x_n\}$ of n objects is partitioned into a set of k clusters $C_i, i = \{1, \dots, k\}$
- Distance is measured to the **centroid** or **medoid** c_i of cluster C_i with

$$\text{centroid} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{medoid} = \arg \min_{y \in X} \sum_{i=1}^n \text{dist}(y, x_i)$$

- Sum of squared distances E is minimized

$$E = \sum_{i=1}^k \sum_{p \in C_i} \text{dist}(p, c_i)^2$$

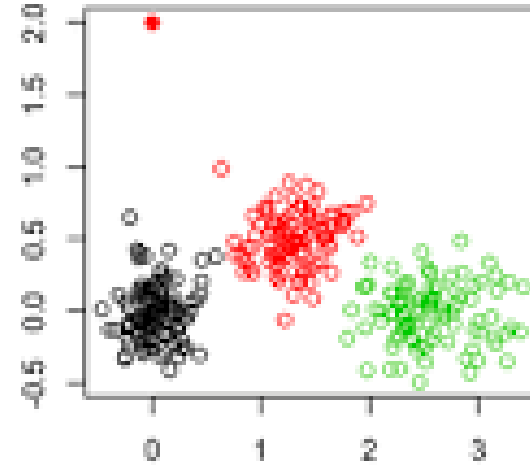
Partitioning criteria

- **Global optimal**: exhaustively enumerate all partitions
- **Heuristic methods**
 - *k-means* (MacQueen'67, Lloyd'57/'82):
Each **cluster** is represented by the **center** of the cluster
 - *k-medoids* or **PAM** (Partition around medoids) (Kaufman & Rousseeuw'87):
Each cluster is represented by **one of the objects** in the cluster

The Problem of the K-Means Algorithm

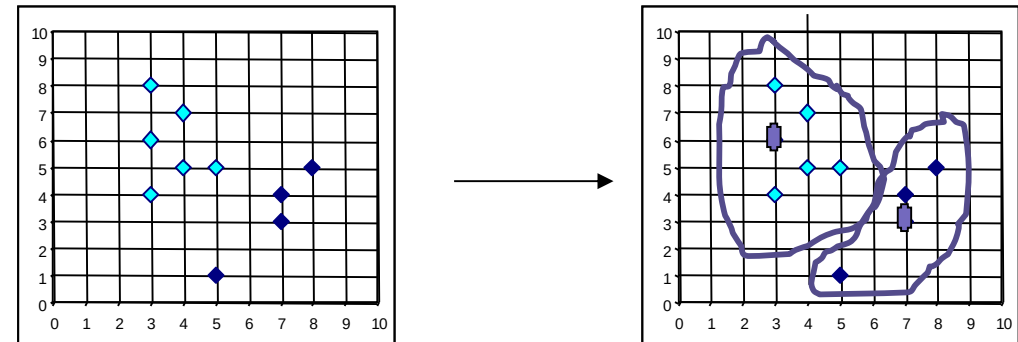
The **k-means** algorithm is sensitive to outliers:

- Object with an extremely large value may substantially distort the distribution of the data



k-medoids:

- Instead of taking the **mean** value of the object in a cluster as a reference point, **medoids** can be used, which is the **most centrally located** object in a cluster



Structure of this Lecture

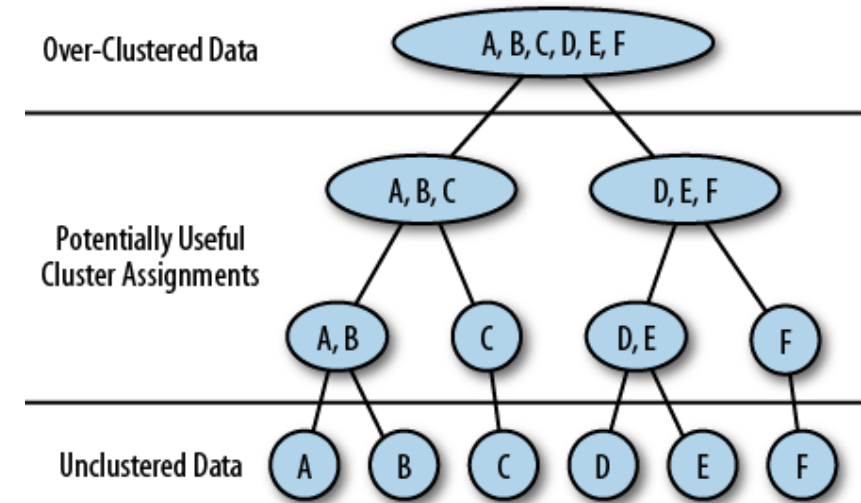
1. Introduction to Machine Learning
2. Similarity and Distances
3. Clustering
 1. Overview
 2. Partitioning
 - 3. Hierarchical clustering**
 4. Density-based
 5. Cluster assessment
 6. Aspects of big data clustering

Hierarchical Clustering – Main Idea

Define distance between clusters

Merge/Split clusters based on distance

Use distance matrix as clustering criteria



A. Kouznetsov, M. Tsvetovat „Social Network Analysis for Startups“, O’Reilly, 2011

Characteristics:

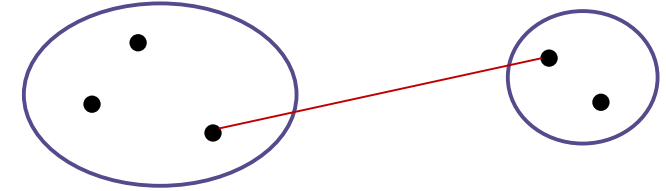
- Hierarchical clustering does not require the number of clusters k as an input
- Hierarchical clustering needs a termination condition

Hierarchical Clustering – Distances between Cluster

Single linkage

distance between the two closest members of each cluster

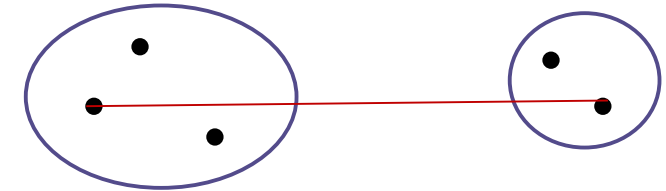
$$d_{SL}(G, H) = \min_{x \in G, y \in H} d(x, y)$$



Complete linkage

distance between the most distant members of each cluster

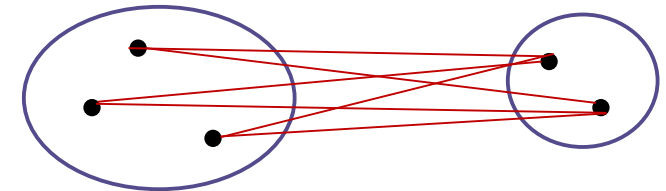
$$d_{CL}(G, H) = \max_{x \in G, y \in H} d(x, y)$$



Average linkage

the average distance between the points from two clusters

$$d_{Avg}(G, H) = \frac{1}{n_G n_H} \sum_{x \in G} \sum_{y \in H} d(x, y), n_G = |G|, n_H = |H|$$



Hierarchical Clustering

**divisive
(DIANA)**

Step 0

Step 1

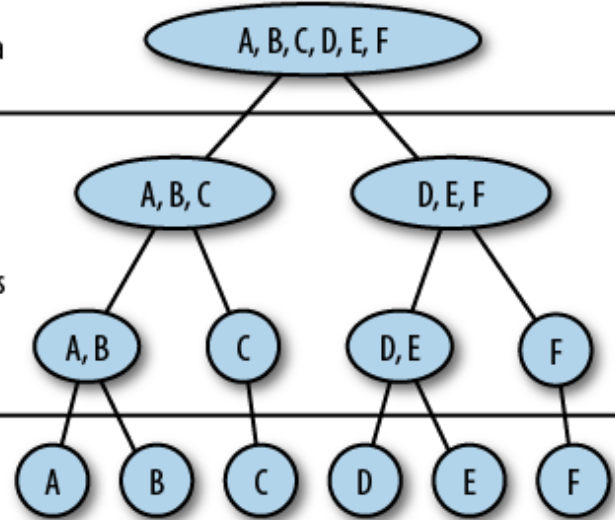
Step 2

Step 3

Over-Clustered Data

Potentially Useful
Cluster Assignments

Unclassified Data



**agglomerative
(AGNES)**

Step 3

Step 2

Step 1

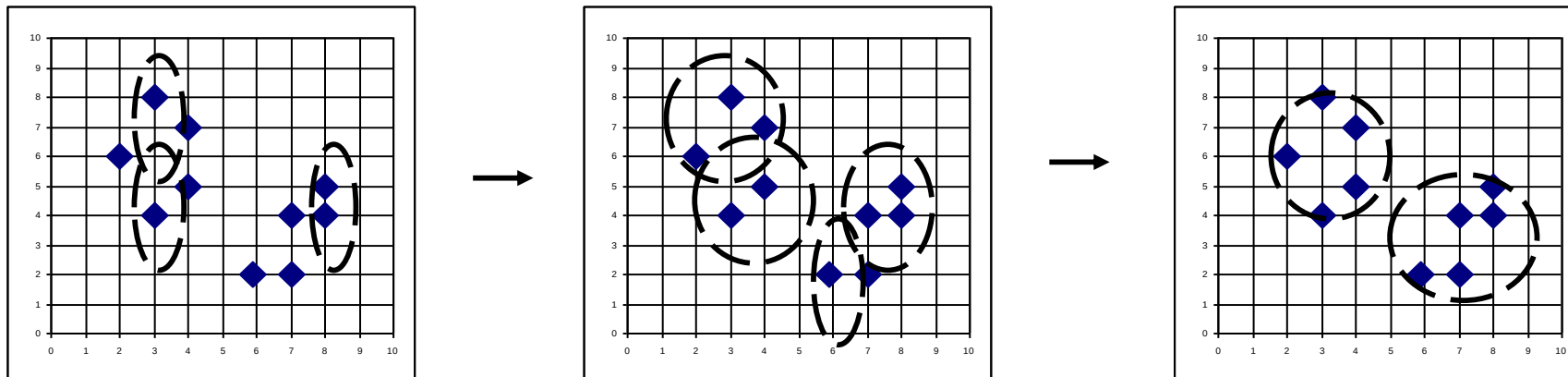
Step 0



adapted from [A. Kouznetsov, M. Tsvetovat „Social Network Analysis for Startups“, O'Reilly, 2011]

AGNES (Agglomerative Nesting)

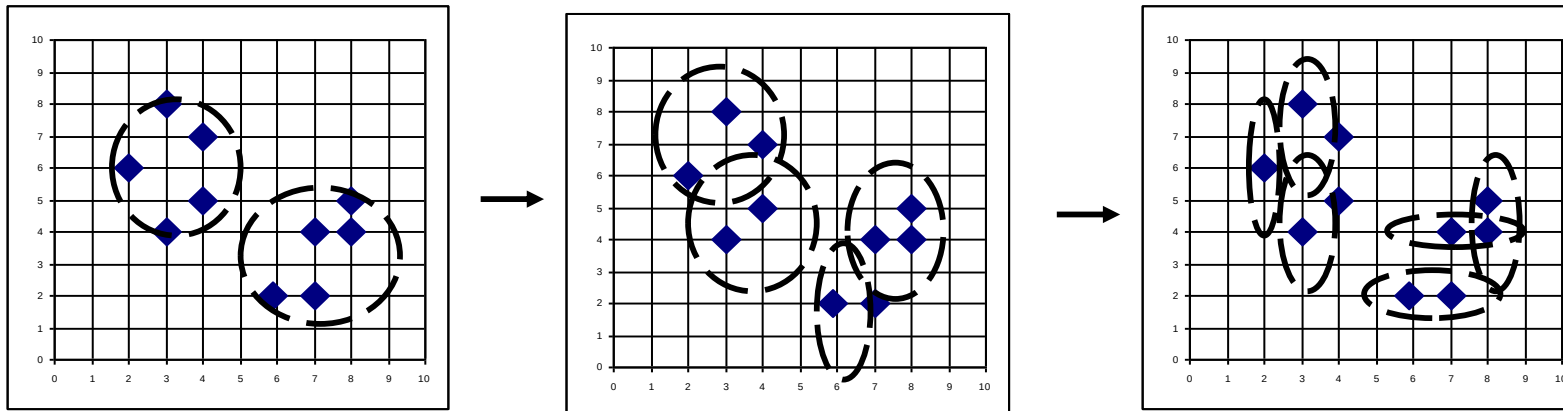
- Introduced 1990 by Kaufman, L. and Rousseeuw, P.*
- Use the **single-link method** and the dissimilarity matrix
- **Merge nodes** that have the **least dissimilarity**
- Eventually all nodes belong to the same cluster
- Implemented in statistical analysis packages, e.g., R



* [Kaufman, L., Rousseeuw, P. (1990). "Finding Groups in Data: An Introduction To Cluster Analysis." 10.2307/2532178]

DIANA (Divisive Analysis)

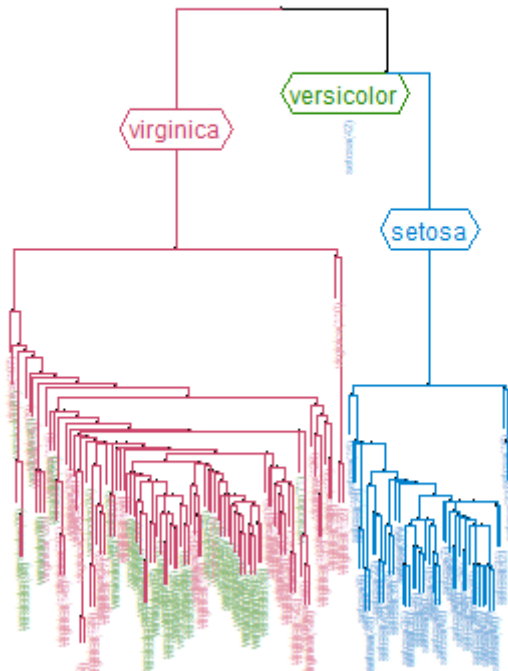
- Introduced 1990 by Kaufman, L. and Rousseeuw, P.*
- **Inverse** order of **AGNES**
- Eventually each node forms a cluster on its own
- Implemented in statistical analysis packages, e.g., R



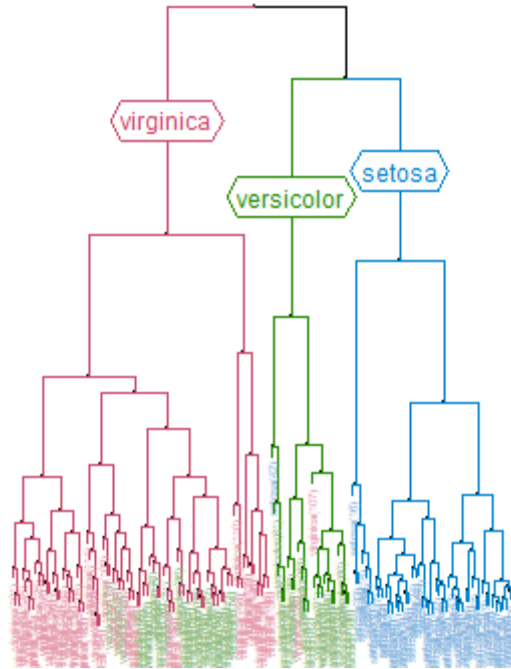
[Kaufman, L., Rousseeuw, P. (1990). "Finding Groups in Data: An Introduction To Cluster Analysis." 10.2307/2532178]

Hierarchical Clustering – Cluster Distance (Iris Data Set)

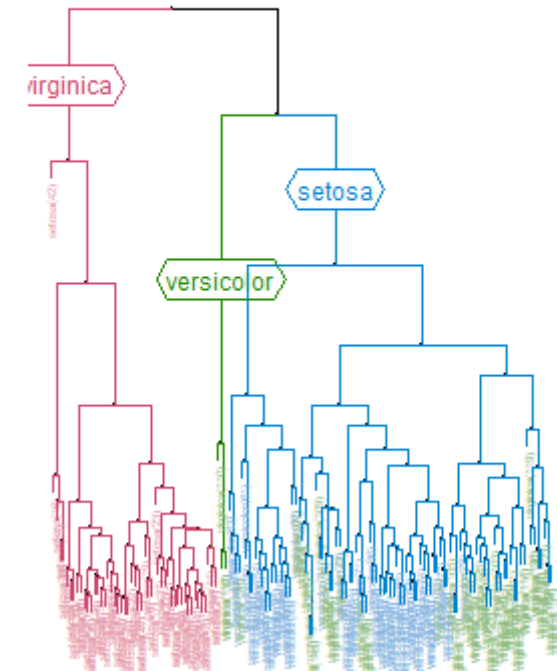
Cluster Dendrogram for iris
distance euclidean
method single



Cluster Dendrogram for iris
distance euclidean
method complete



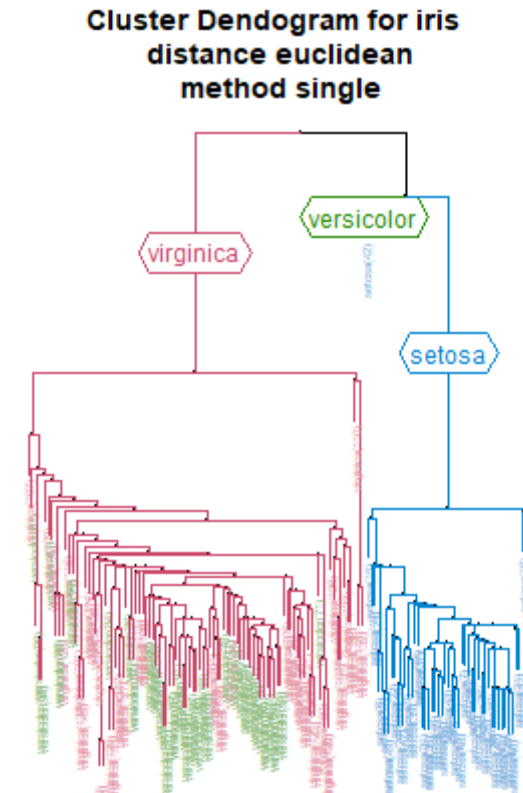
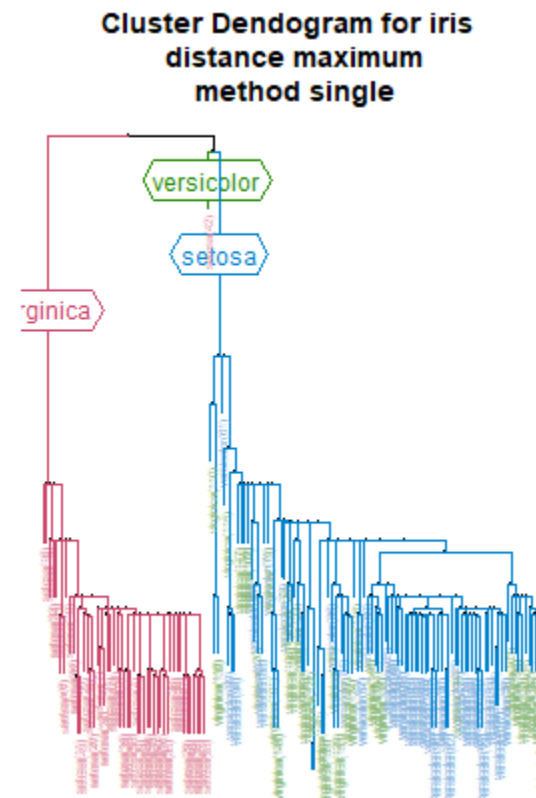
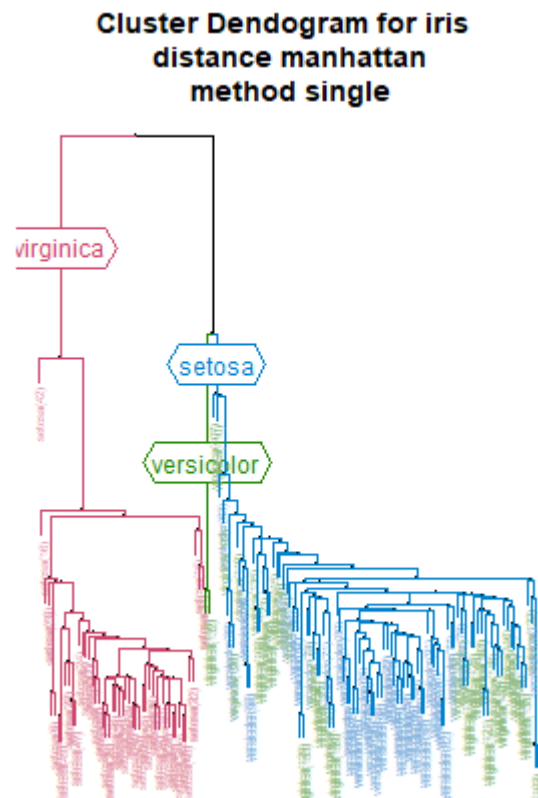
Cluster Dendrogram for iris
distance euclidean
method average



Plots: Courtesy of Iryna Okhrin

Dendrogram: see A.J. Izenman, *Modern Multivariate Statistical Techniques*, Springer Texts in Statistics, Springer, 2008, p.417

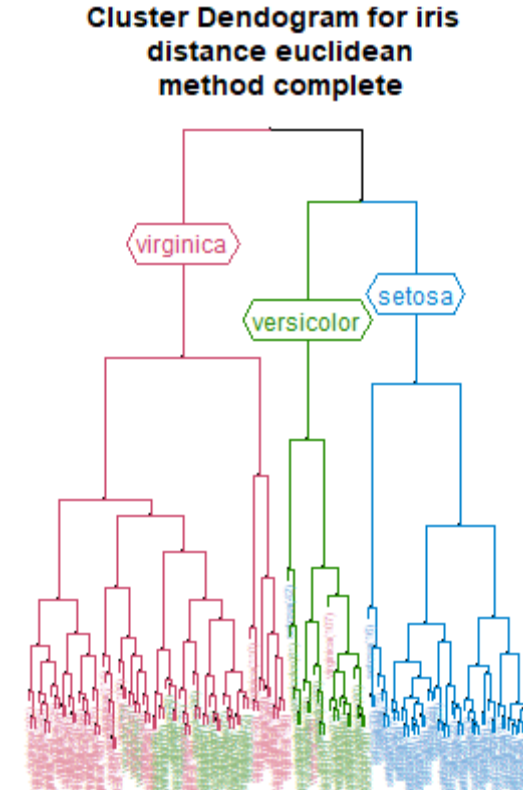
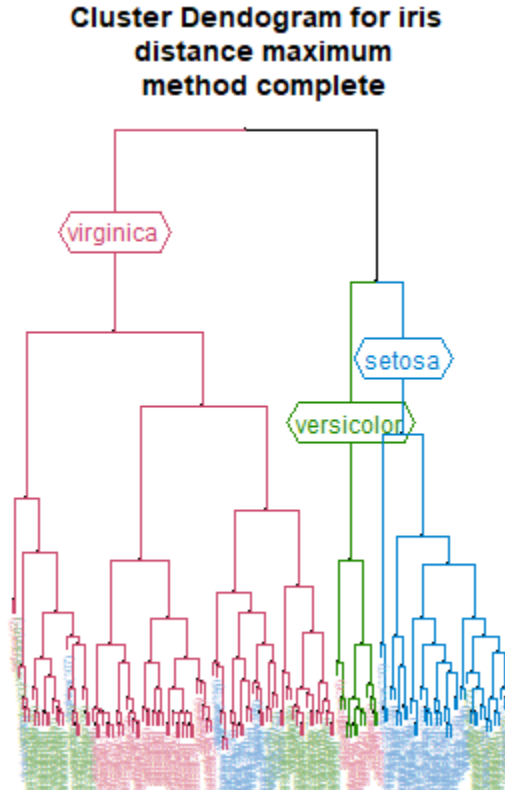
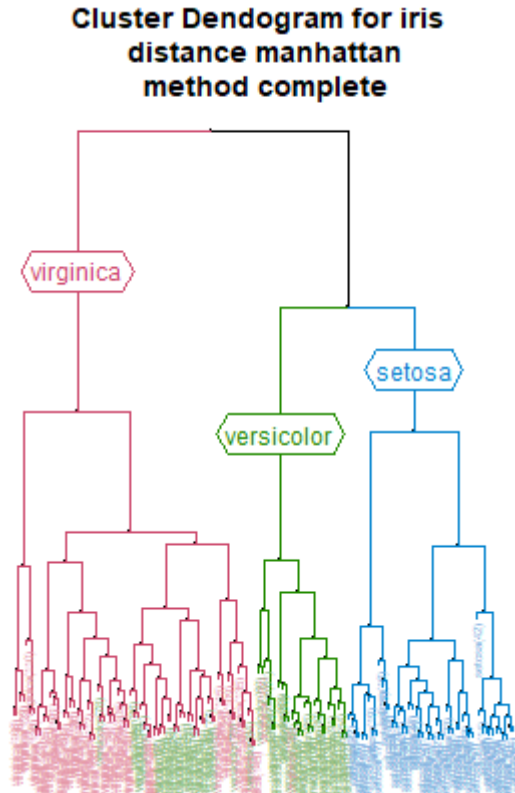
Hierarchical Clustering – Cluster Distance Single (Iris Data Set)



Plots: Courtesy of Iryna Okhrin

Dendrogram: see A.J. Izenman, *Modern Multivariate Statistical Techniques*, Springer Texts in Statistics, Springer, 2008, p.417

Hierarchical Clustering – Cluster Distance Complete (Iris Data Set)



Plots: Courtesy of Iryna Okhrin

Dendrogram: see A.J. Izenman, *Modern Multivariate Statistical Techniques*, Springer Texts in Statistics, Springer, 2008, p.417

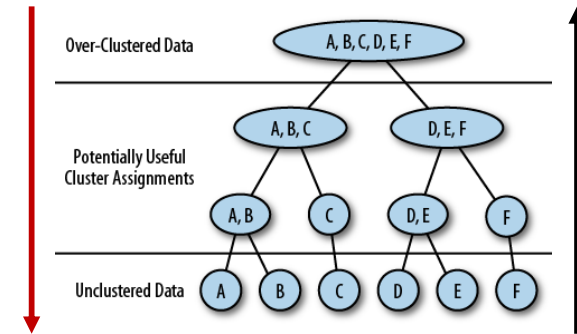
Hierarchical Clustering

Pro:

- **Flexibility** due to usage of complex distance measure
- **No other parameters** than distance function and fusioning method
- Cluster result allows **substructures**

Cons:

- **Analysis** effort of the result
- Cluster analysis provides **many partitions** of the data into clusters
- User must decide **how to partition**

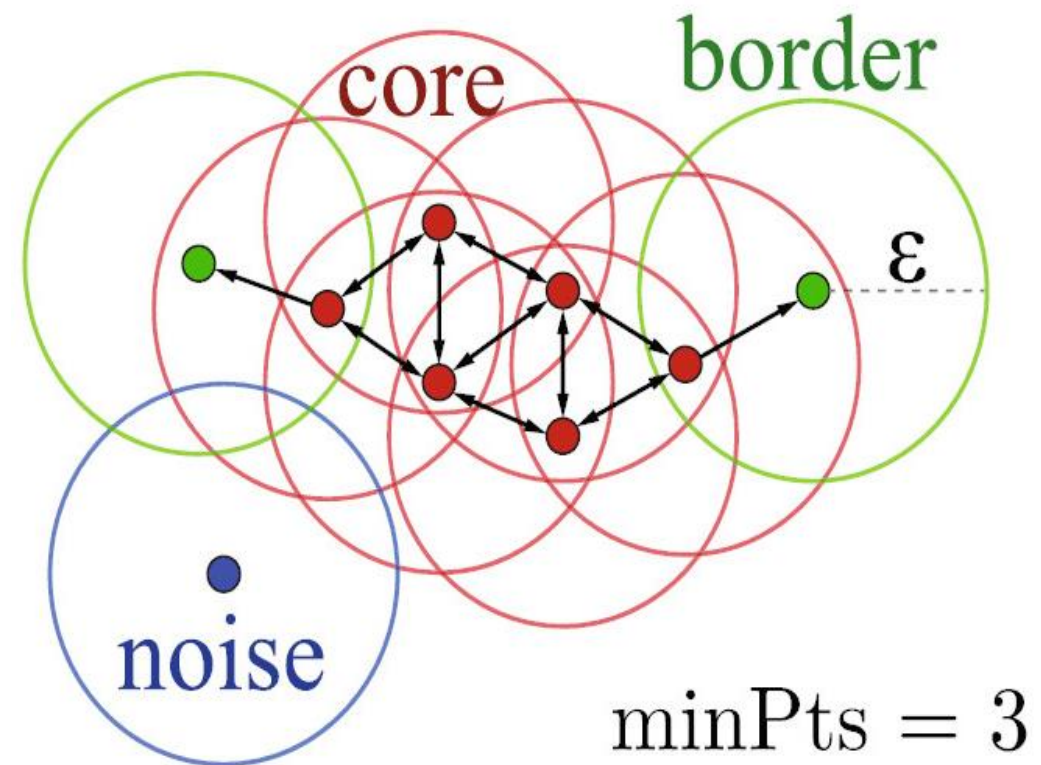


Structure of this Lecture

1. Introduction to Machine Learning
2. Similarity and Distances
3. Clustering
 1. Overview
 2. Partitioning
 3. Hierarchical clustering
 - 4. Density-based**
 5. Cluster assessment
 6. Aspects of big data clustering

Density-based – DBSCAN Algorithm

- Density-Based Spatial Clustering of Applications with Noise [1]
- Parameters
 - ϵ : maximum distance, so that points are reachable
 - minPts: min. number of points within ϵ -distance to define a core point
- Types of points
 - Core: at least minPts points (including the point itself) within distance ϵ
 - Border: connected to a core point, but connected to less than minPts points in distance ϵ
 - Noise: neither core nor border point

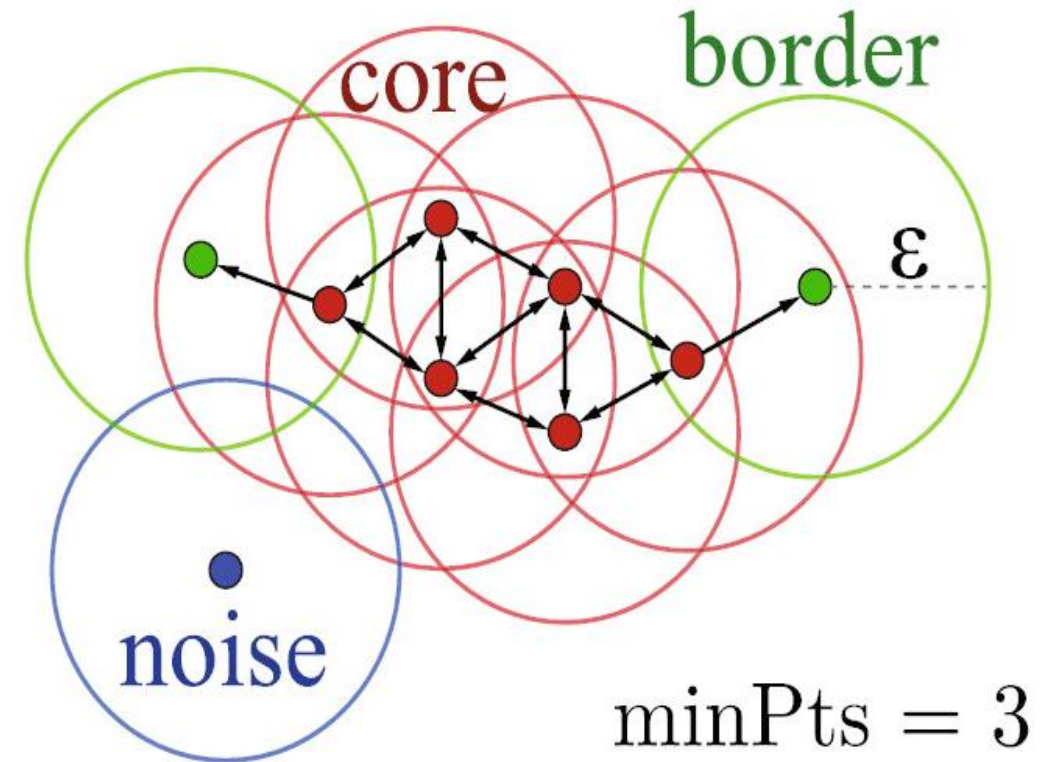


A. Mehle, B. Likar, D. Tomaževič. 2017. In-line recognition of agglomerated pharmaceutical pellets with density-based clustering and convolutional neural network. IPSJ Transactions on Computer Vision and Applications

[1] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu. 1996. A density-based algorithm for discovering clusters in large spatial databases with noise. In Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD'96). AAAI Press, 226–231.

Density-based – DBSCAN Algorithm

```
DBSCAN(D, eps, MinPts)
  C = 0
  for each unvisited point P in dataset D
    mark P as visited
    NeighborPts = D.regionQuery(P, eps)
    if sizeof(NeighborPts) < MinPts
      mark P as NOISE
    else
      C = next cluster
      expandCluster(P, NeighborPts, C, eps, MinPts)
expandCluster(P, NeighborPts, C, eps, MinPts)
  add P to cluster C
  for each point P' in NeighborPts
    if P' is not visited
      mark P' as visited
      NeighborPts' = D.regionQuery(P', eps)
      if sizeof(NeighborPts') >= MinPts
        NeighborPts = NeighborPts joined with NeighborPts'
      if P' is not yet member of any cluster
        add P' to cluster C
        unmark P' as NOISE if necessary
regionQuery(P, eps)
  return all points within P's eps-neighborhood (including P)
```



A. Mehle, B. Likar, D. Tomažević. 2017. In-line recognition of agglomerated pharmaceutical pellets with density-based clustering and convolutional neural network. IPSJ Transactions on Computer Vision and Applications

DBSCAN vs. K-Means

Dataset



DBSCAN



K-Means



Pictures: <https://towardsdatascience.com/understanding-dbscan-algorithm-and-implementation-from-scratch-c256289479c5>

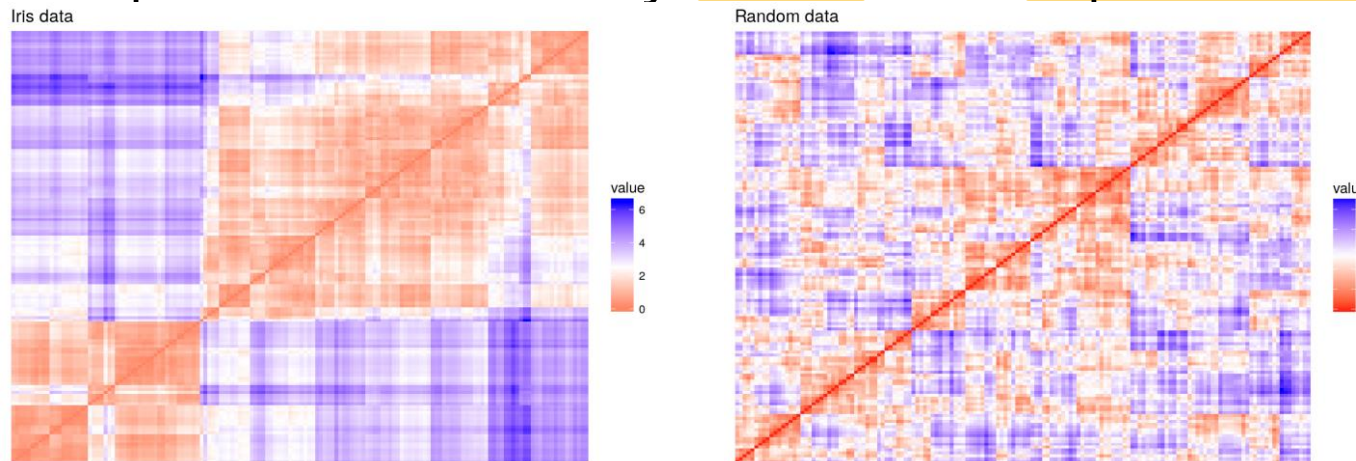
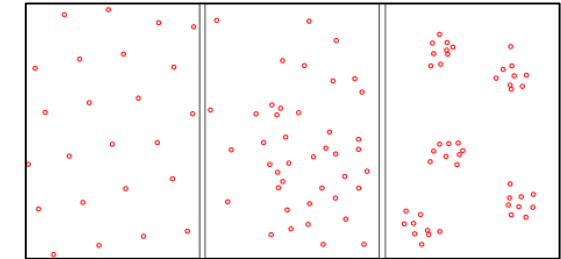
Structure of this Lecture

1. Introduction to Machine Learning
2. Similarity and Distances
3. Clustering
 1. Overview
 2. Partitioning
 3. Hierarchical clustering
 4. Density-based
 - 5. Cluster assessment**
 6. Aspects of big data clustering

Cluster Assessment - Data

Assessing Clustering Tendency

- Assess if non-random structure exists
- **Probability** that data is generated by uniform data distribution
- Test spatial randomness by **statistic** test: **Hopkins Statistic** [1]



Plots from [2]

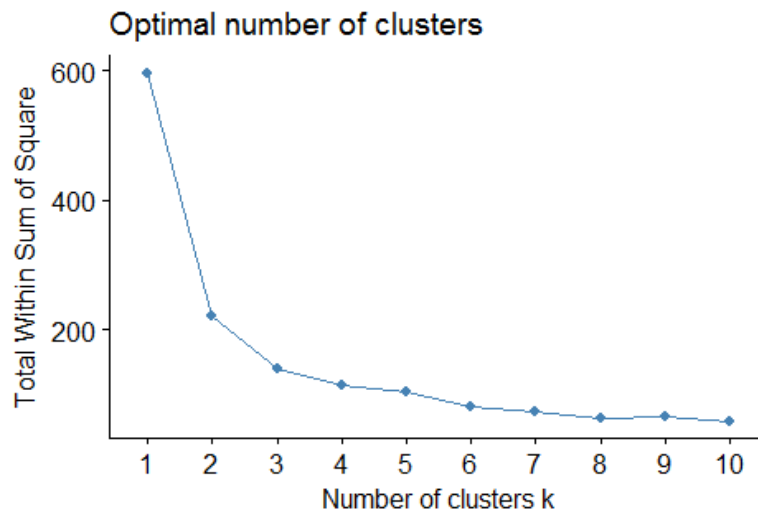
[1] Cross, G.R.; Jain, A.K. (1982). "Measurement of clustering tendency". Theory and Application of Digital Control: 315-320. doi:10.1016/B978-0-08-027618-2.50054-1.

[2] <https://www.kaggle.com/code/meran2k/clustering-evaluation-of-iris-dataset/report>

Cluster Assessment – Determining the Number of Clusters

Elbow Method

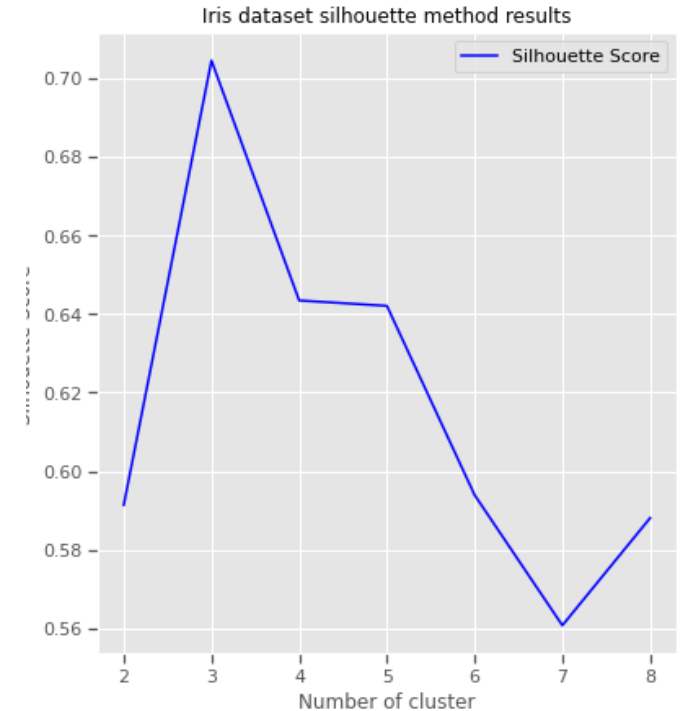
- Compute the sum of squared distances for each k
- Optimal number of clusters: sharp turn in the Elbow graph
- Very subjective as it is often ambiguous where the elbow is



Cluster Assessment – Determining the Number of Clusters

Silhouette Method

- Investigate separation and compactness of clusters
- **Silhouette Coefficient** $sc(i)$ for each data point i :
 - $sc(i) = \frac{(b_i - a_i)}{\max(a_i, b_i)}$
 - a_i : mean distance to the other points in the same cluster, measure of compactness
 - b_i : mean distance to other points in the next closest cluster, measure of separation
 - $-1 \leq sc(i) \leq 1$
- **Silhouette score** of a clustering K defined as mean silhouette coefficient over all instances.



Cluster Assessment - Methods

External evaluation

- „Ground truth“ required, e.g. labels in Iris data set
- **Purity**, see [1]
- **Jaccard Index** $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$
- **F-score**, **Confusion Matrix**, see lecture supervised learning

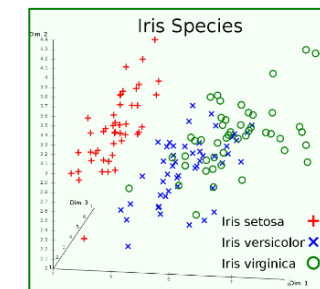
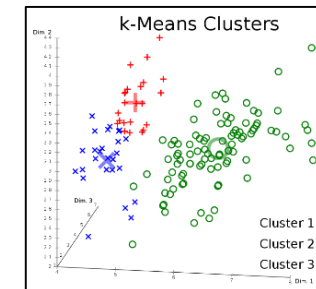
Manual evaluation

- assessing by humans

Indirect evaluation

- assessing by evaluating the application
- [1] C.P. Murphy. Probabilistic machine Learning. MIT Press, 2022, p. 712

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
4.9	3.1	1.5	0.1	setosa
5.1	3.8	1.5	0.3	setosa
4.7	3.2	1.6	0.2	setosa
5.1	3.4	1.5	0.2	setosa
5.0	3.3	1.4	0.2	setosa
5.2	2.7	3.9	1.4	versicolor
5.7	2.6	3.5	1.0	versicolor
5.7	2.8	4.1	1.3	versicolor
6.0	2.2	5.0	1.5	virginica



Structure of this Lecture

1. Introduction to Machine Learning
2. Similarity and Distances
3. Clustering
 1. Overview
 2. Partitioning
 3. Hierarchical clustering
 4. Density-based
 5. Cluster assessment
 6. **Aspects of big data clustering**

Aspects of Big Data Clustering

Challenge: Scalability

- clustering algorithms often with high time complexity (e.g. k-means NP-hard)
- Speed up/scale up clustering algorithms with minimum sacrifice to clustering quality

Procedure:

- Initialization (e.g. randomly choose k cluster centers)
- iterative process until some convergence criteria (e.g., assign each data point to closest cluster centers in k-means)

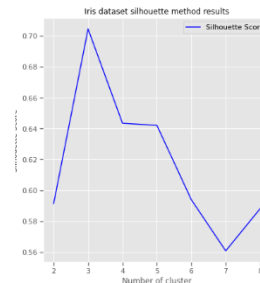
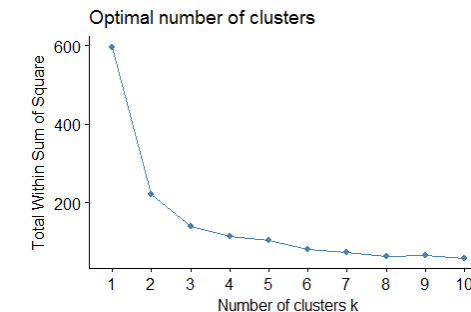
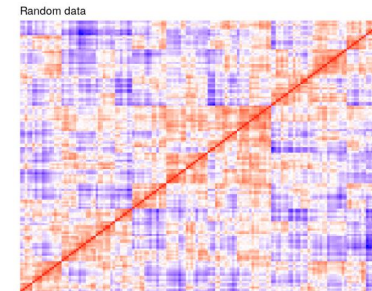
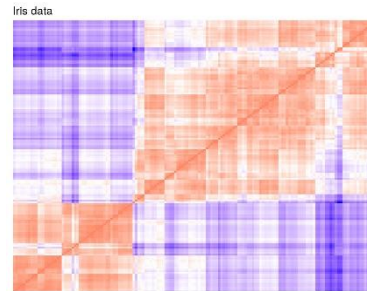
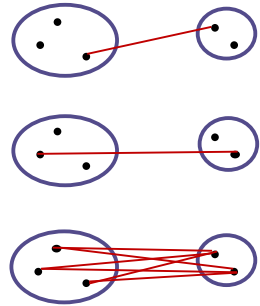
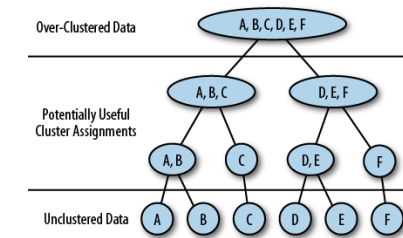
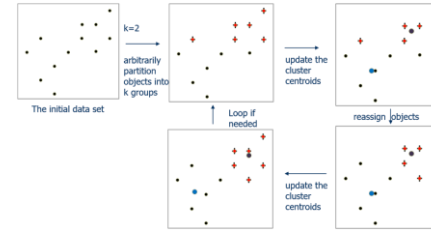
Speed up/Scale up:

- reducing the iteration number
- reducing access to data points (e.g. randomized techniques)
- distributing/parallelizing computation

[Aggarwal, C., Reddy K. (eds): Data Clustering: Algorithms and Applications, Chapman and Hall/CRC, 2014]

Clustering Takeaway

- Types of clustering
 - partitioning (K-Means)
 - hierarchical clustering
 - density (DBSCAN)
- Clustering depends on distances
- Assessment is a challenge
 - without ground truth
 - with ground truth
- Clustering large data
 - diminish set of data
 - distributed computing
 - high dimensional data are different (curse of dimensionality)



Today's Lecture

- Introduction Machine Learning
 - Unsupervised Learning
 - Supervised Learning
 - Similarities and Distances
- Clustering
 - Partitioning
 - Hierarchical clustering
 - Density-based
 - Assessment of clustering results
 - Applications

Further Reading

Data Mining

Han, J., Kamber, M., Pei, J.: *Data Mining - Concepts and Techniques*, Morgan Kaufmann, 2012, chapter 10/11
http://hanj.cs.illinois.edu/bk3/bk3_slidesindex.htm

Zaki, Mohammed J. and Meira, Wagner, Jr: *Data Mining and Machine Learning: Fundamental Concepts and Algorithms*, Second Edition, Cambridge University Press, March 2020, <https://dataminingbook.info/>

Machine Learning

Murphy, K.P. *Probabilistic Machine Learning*, MIT Press, 2022, via SLUB electronically available

James, G., Witten, D., Hastie, T., Tibshirani, R. *An Introduction to Statistical Learning*, Springer 2021, <https://www.statlearning.com/>

A.J. Izenman, *Modern Multivariate Statistical Techniques*, Springer Texts in Statistics, Springer, 2008
http://weixingsong.weebly.com/uploads/7/4/3/5/7435707/modern_multivariate_analysis.pdf

Shalev-Shwartz, S., Ben-David, S. *Understanding Machine Learning: From Theory to Algorithms*, 2014
<https://www.cambridge.org/core/books/understanding-machine-learning/3059695661405D25673058E43C8BE2A6>

Digitization and Data Analytics

Machine Learning I

Types of Machine Learning

Semi-Supervised Learning

- Use both, labeled and unlabeled data
- Chapelle, O., Schölkopf, B., Zien, A. (2006). Semi-Supervised Learning. MIT Press.
- Ouali, Y., Hudelot, C., Tami, M. (2020). An Overview of Deep Semi-Supervised Learning
<https://arxiv.org/pdf/2006.05278.pdf>

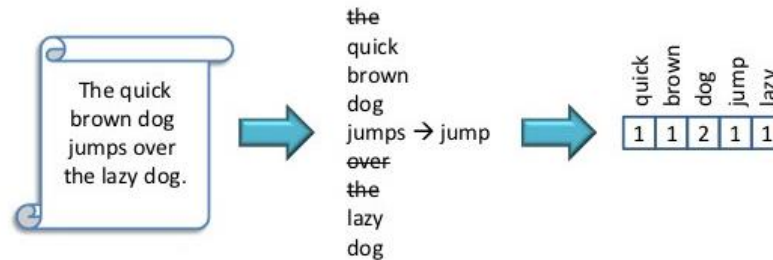
Reinforcement Learning

- Learning by occasional reward or punishment signals
- Examples: a baby learns to walk, dog training
- Richard S. Sutton and Andrew G. Barto. 1998. Introduction to Reinforcement Learning (1st ed.). MIT Press, Complete draft second edition March 2018 <http://incompleteideas.net/book/bookdraft2018mar11.pdf>
- Will not be topic of this lecture series!

Measure the Quality of Clustering

Dissimilarity/Similarity metric

- similarity is expressed in terms of a distance function $d(\vec{x}, \vec{y})$, typically metric
- definition different for interval-scaled, boolean, categorical, ordinal ratio, and vector variables
- definition dependant on applications and data semantics
- Example:



Goal: separate sentences about „dog“ from others

feature: „dog“

added weight to feature „dog“ influences distance to other sentences

Quality of clustering

- separate “quality” function measures “goodness” of a cluster
- hard to define “similar enough” or “good enough”

Hierarchical Clustering - Complexity

In general

$O(N^3)$ time, with N : number of data points

Can be reduced to

$O(N^2 \log(N))$ time, see [1]

Single link

$O(N^2)$ time, see [2]

Machine Learning – The Learning Perspective

Schutzraum

„Ever since **computers** were invented, we have wondered whether they **might be made to learn**. If we could **understand** how to program them to learn-to improve automatically with experience-the impact would be dramatic. [...] and a theoretical understanding of learning is beginning to emerge. ”

Thomas M. Mitchell. *Machine Learning* (1 ed.). McGraw-Hill, Inc., New York, NY, USA, **1997**

A computer program is said **to learn from experience E** with respect to some class of **tasks T**,

Schutzraum

and **performance measure P**, if its performance at tasks in T, as measured by P, **improves**

Machine Learning – A Probabilistic Perspective

Schutzraum

“Roughly speaking, this means that we **treat all unknown quantities** (e.g., predictions about the future value of some quantity of interest, such as tomorrow’s temperature, or the parameters of some model) as **random variables**, that are endowed **with probability distributions** which describe a weighted **set of possible values** the variable may have.”

Murphy, K.P. *Probabilistic Machine Learning: An introduction* MIT Press, 2022

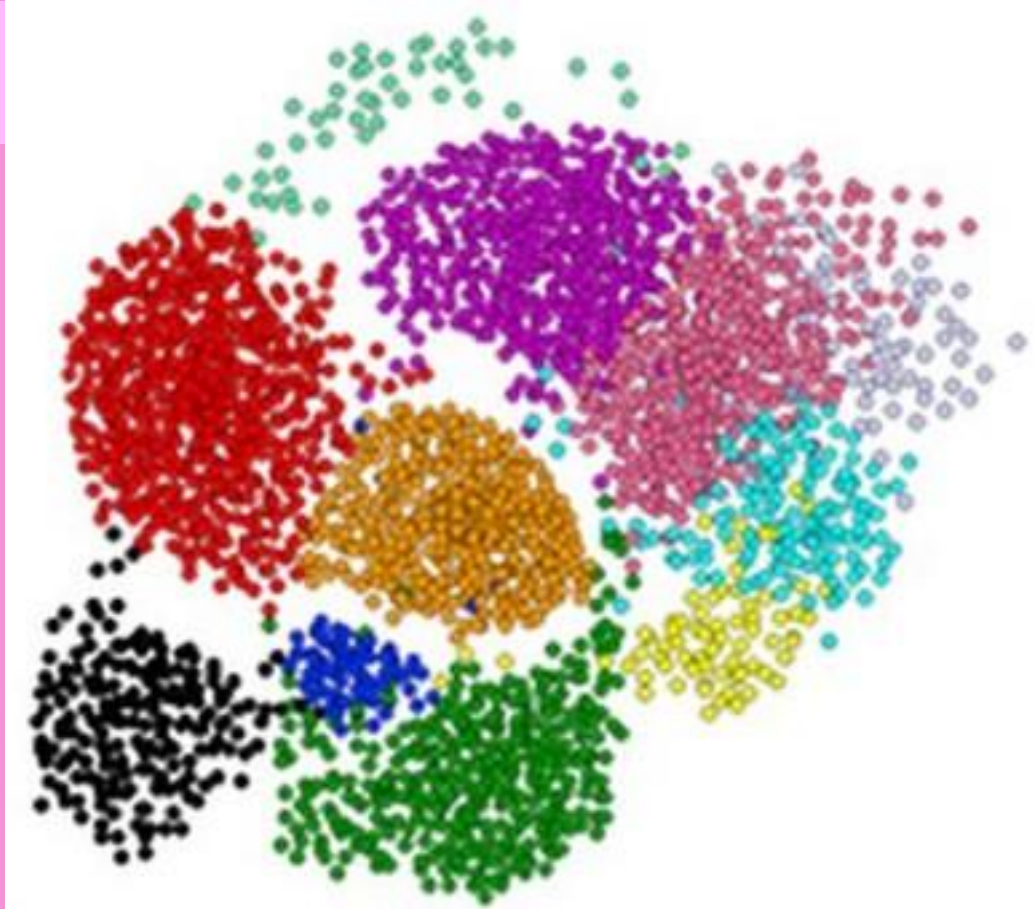
Schutzraum

Types of Machine Learning – Types of Data

Schutzraum		
	<i>Supervised Learning</i>	<i>Unsupervised Learning</i>
<i>Discrete</i>	classification	clustering
<i>Continuous</i>	regression	dimensionality reduction

Schutzraum

Clustering - A Hard Problem



Schutzraum